Title of Your Ph.D. Thesis

Your Name

A Thesis Submitted in Partial Fulfillment

of the Requirements for the Degree of

Doctor of Philosophy

in

Systems Engineering and Engineering Management

The Chinese University of Hong Kong

January 2021

Thesis Assessment Committee

Professor AAA (Chair)

Professor BBB (Thesis Supervisor)

Professor CCC (Committee Member)

Professor DDD (External Examiner)

Abstract

True Wireless Stereo (TWS) earphones are a key type of head-mounted wearable (HMW), which also includes VR/AR headsets and smart glasses. These smart devices are designed to be worn on the ears or head. Popular products like Apple AirPods have made earphones the largest segment in the wearable device market, with global shipments expected to surpass 18 billion units in market value by 2024. Typically, there are two main types of sensors on earphones: microphones and inertial measurement units (IMUs). Note that these technologies are commonly used in all head-mounted wearables, not just earphones. The sensing abilities of earphones are often limited because manufacturers prioritize delivering content, like spatial audio and 3D video. Therefore, this dissertation aims to enhance the sensing capabilities of earphones. Unlike existing research that typically focuses on either the environment or the user, our work uniquely models both. Furthermore, it is essential to consider the interactions, or coupling, between humans and their environment. By achieving a comprehensive understanding of this human-environment coupling, earphones can become a gateway to a smarter lifestyle for users.

This dissertation specifically introduces *VibVoice*, the first system designed for speech enhancement on existing head-mounted wearables (HMWs). Environmental noise can significantly interfere with voice-related applications on these devices, which represents one of the key interactions between humans and their surroundings. The compact design of HMWs presents significant challenges for existing solutions. To address this, we propose utilizing the bone-conduction effect of the user's voice to filter out background noise. The principle behind VibVoice is that the inertial measurement unit (IMU), which is commonly

found in head-mounted wearables, can detect the user's clean voice through vibrations on the head. Hence, we design a two-branch encoder-decoder deep neural network to fuse the high-level features of the two modalities and reconstruct clean speech. To address the issue of insufficient paired data for training, we extensively measure the bone conduction effect from a limited dataset to extract the physical impulse function for cross-modal data augmentation. We evaluate VibVoice on multiple HMWs under diverse noises collected in the real world that outperform the two state-of-the-art baselines by a 52% lower word error rate.

Then we try to extend the sensing capabilities of earphones to not only the vocal activity (speech). We propose *EmbodiedSense*, a new earphone sensing system that leverages off-the-shelf sensors to recognize human activity without an unlimited number of classes. The design principle of EmbodiedSense is based on extracting extra information to assist in activity recognition. Firstly, we propose to recognize the long-term scenario that can benefit the activity recognition by clustering the relevant activities. Secondly, we incorporate sound localization to differentiate the user's sound (not only speech) and environmental noises, considering the natural head movement as well. Given the above knowledge, EmbodiedSense employs LLM to process sound and motion events to reason human activities effectively. EmbodiedSense outperforms other multi-modal LLM baselines with more than 20% in our self-collected dataset as well as the public dataset.

Finally, we scale up our system into a multi-device collaborative paradigm. Specifically, we propose *CoHear* that enhances speech at the conversation level, which requires a holistic modeling of all the members in the conversation and an effective way to extract the speech from the mixture. CoHear makes two key contributions: 1) constructs a conversation network automatically, and 2) collaboratively extracts conversations. Specifically, we consider each earphone as a sensor node and estimate the network geometry by observing the natural conversation. Besides, our speech extraction model can leverage the transmitted audio to isolate the relevant part. CoHear is evaluated in both real-world experiments and simulations with a more than 90% accuracy in group formation, improving the speech quality by up to 8.8 dB over state-of-the-art baselines. In a user study with 20 participants, CoHear has a much higher score than baseline, with good usability.

In summary, this dissertation proposes three different systems on HMWs with existing sensors to understand the user, environment, and the coupling between them. These systems provide not only detailed system implementation but also bring new insights into the mobile sensing community.

摘要

真无线立体声（TWS）耳机是头显可穿戴设备（HMW）的一种重要类型，其他类型还包括虚拟现实/增强现实（VR/AR）头显和智能眼镜。这些智能设备设计为佩戴在耳朵或头部。以苹果AirPods为代表的热门产品使耳机成为可穿戴设备市场中最大的细分市场，预计到2024年全球出货量市场价值将超过180亿美元。

通常，头显可穿戴设备采用两种主要的感知技术：麦克风和惯性测量单元（IMU）。这些技术不仅限于耳机，而是广泛应用于所有头显可穿戴设备。然而，耳机的感知能力常常受限，因为制造商更专注于内容传递，例如空间音频和3D视频。因此，本论文旨在提升头显可穿戴设备的感知能力。

与现有耳机感知研究不同，我们的工作独特地同时建模环境和用户，而非仅关注其中之一。此外，考虑人类与环境（包括其他个体）之间的交互（耦合）至关重要。

本论文具体介绍了 VibVoice，这是首个为现有头显可穿戴设备设计的语音增强系统。环境噪声会显著干扰这些设备上的语音相关应用，这是人类与环境交互的关键方面之一。头显可穿戴设备的紧凑设计为现有解决方案带来了重大挑战。为此，我们提出利用用户语音的骨传导效应来滤除背景噪声。VibVoice 的原理是利用头显可穿戴设备中常见的惯性测量单元（IMU）通过头部振动检测用户的干净语音。因此，我们设计了一个双分支编码器-解码器深度神经网络，融合两种模态的高级特征并重建干净语音。为了解决训练数据不足的问题，我们从有限数据集中广泛测量骨传导效应，提取物理冲激函数用于跨模态数据增强。我们在多种头显可穿戴设备上评估了 VibVoice，在现实世界中收集的多种噪声环境下，其表现优于两个最先进的基线，词错误率降低了52%。

随后，我们尝试扩展耳机的感知能力，不仅限于语音活动（说话）。我们提出了 EmbodiedSense，一种利用现成传感器识别人类活动的新型耳机感知系统，且不限于固定数

量的类别。EmbodiedSense 的设计原理基于提取额外信息以辅助活动识别。首先，我们提出通过聚类相关活动来识别长期场景，从而提升活动识别效果。其次，我们结合声音定位技术，区分用户的声音（不仅是语音）和环境噪声，同时考虑自然的头部运动。基于上述知识，EmbodiedSense 利用大语言模型（LLM）处理声音和运动事件，有效推理人类活动。在我们自收集的数据集以及公开数据集上，EmbodiedSense 的表现优于其他多模态大语言模型基线，准确率提高了20%以上。

最后，我们将系统扩展为多设备协作范式。具体来说，我们提出了 CoHear，它在对话层面增强语音，需要对对话中所有成员进行整体建模，并有效从混合音频中提取语音。CoHear 做出了两个关键贡献：1）自动构建对话网络；2）协作提取对话内容。具体而言，我们将每只耳机视为一个传感器节点，通过观察自然对话估计网络几何形状。此外，我们的语音提取模型可以利用传输的音频隔离相关部分。CoHear 在现实世界实验和仿真中进行了评估，在群体形成方面准确率超过90%，语音质量比最先进的基线提高了高达8.8分贝。在一项包含20名参与者的用户研究中，CoHear 的得分远高于基线，表现出良好的可用性。

总之，本论文提出了三种基于头显可穿戴设备现有传感器的系统，以理解用户、环境及其之间的耦合。这些系统不仅提供了详细的系统实现，还为移动感知社区带来了新的见解。

Contents

List of Tables

List of Figures

Acknowledgments

Acknowledgement here.

Chapter 1

Introduction

1.1 Contributions

Chapter 2

Literature Review

Chapter 3

VibVoice

3.1 Introduction

Head-mounted wearables, or HMWs, are smart devices that can be put on users' heads or ears, which include True Wireless Stereo (TWS) earphones, VR/AR headsets, and smart glasses. HMWs are equipped with several sensors and run various applications like VR/AR, motion recognition, and voice assistants. Represented by TWS earphones like the Apple Airpods series, the shipment of HMWs has grown as the largest category among all wearable devices, estimated to be more than 273 million units worldwide in 2023 (Statista, 2020). Manufacturers are adding various functions to HMWs. For example, some HMWs (especially earphones and headphones) support active noise cancellation (ANC) to improve the listening experience. On the speech side, voice-related applications using HMW s microphones are among the most frequently used.

For example, an increasing number of people make phone calls with TWS earphones. Through voice commands, users can also interact with voice assistants (e.g., Siri and Alexa). However, the speech quality on HMWs is unsatisfactory due to the following challenges: First, most HMWs adopt omnidirectional microphones that can receive audio from any angle, leading to extra environmental noises. Second, although most HMWs are equipped with multiple microphones to form an array and remove noises based on beamforming,

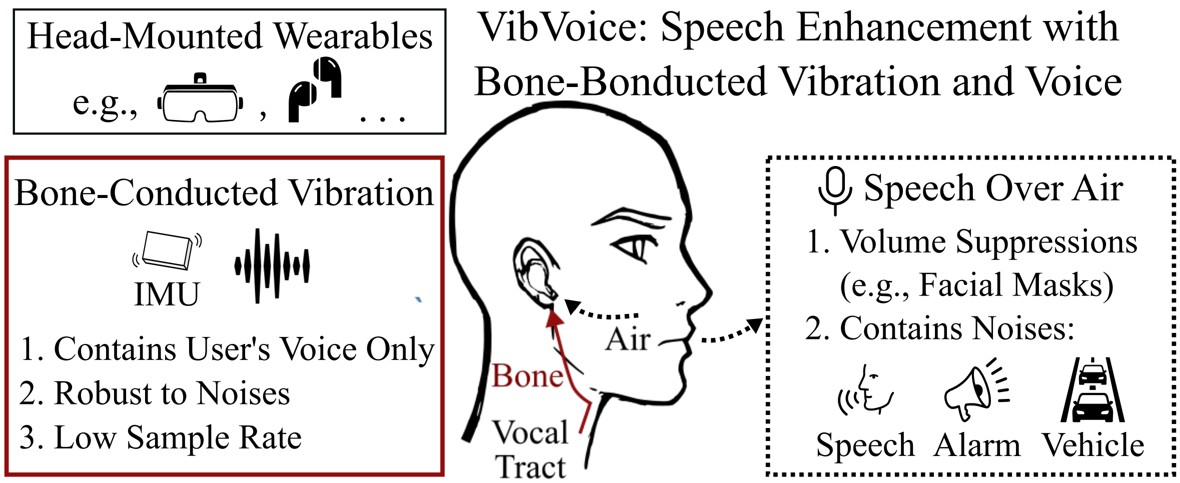


Figure 3.1: *VibVoice enhances speech quality of head-mounted wearables by extracting the user's clear voice from the bone-conducted vibrations.*

the microphones on HMWs are too close to achieving a satisfying performance. Third, the speech audio is significantly suppressed when it reaches the HMWs as the HMWs are usually far away from the user's mouth. Lastly, the worldwide pandemic forces people to wear masks, which can further reduce the audio intensity.

Various approaches have been developed for speech enhancement. Signal processing-based methods (Ephraim and Van Trees, 1995) remove noises based on their statistical models. However, these approaches cannot handle complex environments. Microphone beamforming (Yousefian et al., 2014; Ji et al., 2017) removes noises based on their directions. But they fail to distinguish the speaker's voice and noise when the microphones are placed too close. Several research (Subakan et al., 2021; Zhang and Li, 2021; Chatterjee et al., 2022) adopt deep neural networks (DNNs) to improve speech quality. However, their performance varies when the domain changes. Except for the audio-only solution, several works leverage other modalities like contact sensors (Gupta et al., 2018), vibration sensor (Maruri et al., 2018), camera (Gao and Grauman, 2021), mm-Wave Radar (Ozturk et al., 2021; Li et al., 2020; Liu et al., 2021), ultrasonic sensing (Sun and Zhang, 2021; Zhang et al., 2021), and Lidar (Sami et al., 2020), which introduce additional hardware requirements or user overhead to existing HMWs.

Inertial Measurement Unit (IMU) is a common sensor on most HMWs and can capture audio with a direct connection with the speaker (Ba et al., 2020; Wang et al., 2021). Since HMWs are well connected to the user's head, the IMU accelerometer can sense the vibrations caused by the speaker's voice through the skull's bone conduction. The bone-conducted vibration contains speech from the user only without environmental noises or voices. However, the accelerometer on HMWs has a low sample rate (e.g., 1.6 kHz) due to the limited size and energy consumption, which is not enough to recover the human's voice (e.g., up to 3.4 kHz (Wikipedia, 2020b)).

We propose *VibVoice*, the first end-to-end multi-modal speech enhancement system for HMWs using the bone conduction from the vibration to the audio. The design of VibVoice fits mobile devices, considering the on-device sensor, execution latency, and communication overhead. VibVoice exploits the complementary characteristics of the two modalities: the acceleration has a low sample rate covering a partial audible spectrum but with no environmental noises. The microphone has a high sample rate covering the whole audible spectrum but is mixed with ambient voice and noises. Such a combination of modalities provides a stable reference for extracting the target speech. VibVoice uses a DNN to reconstruct clean speech with two encoders and decoders for processing the data from two modalities, respectively. VibVoice faces three major technical challenges as follows:

Lack of paired labeled acceleration and audio. Collecting a paired multi-modal dataset is labor-intensive. Although there are public datasets of either acceleration or audio, none has the paired data on head-mounted wearables. Collecting such a paired dataset on a large scale can introduce excessive overheads, such as recruiting diverse volunteers and recording hundreds of hours of audio with annotations.

Fusion of data with different modalities and sample rates. Microphone audio has much richer information than acceleration data due to the higher sample rate. As a result, the DNN is prone to overfitting and may ignore the data input of acceleration. Therefore, during the training of DNN, we require a special training strategy to avoid overfitting and two dedicated losses to balance the weights of the two modalities.

Practical concern for deployment. Deploying deep learning speech enhancement in the real world is non-trivial due to the variance among people's voice and bone conduction channels. Even though some users may tolerate a rapid data collection phase before use, the amount and quality of the collected data are not guaranteed for training a high-performance model.

To overcome the above challenges, we design VibVoice based on our extensive measurements of the bone-conduction effects on real-world data. We conclude our contributions as follows:

- We propose a novel estimation approach to model Bone Conduction Function with a limited dataset to augment paired acceleration and audio based on a large public audio dataset. The augmented paired dataset can be used for model training.
- We design a two-branch DNN that a) recovers the speech-related information from the acceleration signal with a low sample rate, b) extracts the clean speech from contaminated audio with the help of the acceleration signal, and c) has a lightweight design and can be run on the mobile device with low latency.
- We collect a two-modal dataset and evaluate VibVoice on a self-designed platform, *EarSense*, equipped with an accelerometer and a microphone to acquire paired acceleration and audio data simultaneously. The dataset and the design of data collector are open-sourced.

We evaluate VibVoice on a paired acceleration and microphone from 15 volunteers. We test VibVoice's performance on the offline synthetic noise dataset and the concurrent noise dataset in various scenarios. VibVoice achieves the best performance with 31 times less latency on mobile devices than the two strong baselines. In addition, data augmentation can reduce the requirement of paired data by more than 72 times. We recruit 35 volunteers for a user study in which VibVoice is preferred by 87% users compared to the baseline.

3.2 Related work

Speech enhancement is an essential function in voice communication that has received extensive attention. We categorize existing speech enhancement methods based on the input modalities, i.e., *audio-only*, *multi-modal*, as well as works sensing acoustic signals with other modalities.

3.2.1 Audio-only Speech Enhancement

Model-driven

Traditional speech enhancement either relies on assumptions of stationarity of signals, independence of speech, noise in the time-frequency domain (Ephraim and Van Trees, 1995), or multiple omnidirectional microphones (e.g., a microphone array) to improve audio quality. The microphone array leverages the difference of arrival times (known as beamforming) toward each microphone to determine the direction of the speaker for speech enhancement (Yousefian et al., 2014; Ji et al., 2017), and speech separation (Wood et al., 2017; Kitamura et al., 2016). However, those approaches cannot adapt to dealing with dynamic noises without prior knowledge.

Machine Learning

Recent studies adopt DNN (Subakan et al., 2021; Zhang and Li, 2021; Chatterjee et al., 2022) for speech enhancement. The deep structure of neural networks can capture the inherited features of the target voice and potential noises based on a large training dataset. DNN-based methods work on both single-channel audio (Subakan et al., 2021) and multi-channel audio with arbitrary microphones layouts (Zhang and Li, 2021). ClearBuds (Chatterjee et al., 2022) uses a DNN to process the stereo audio recorded by customized earbuds, which causes excessive communication overhead. ClearBuds filters noises based on the angles of sources, which does not work when a competing speaker stands at the same angle as the speaker. Although audio-only deep learning speech enhancement achieves good performance, they rely on the training data domain and fail to enhance the speech quality with slim-volume audio.

3.2.2 Multi-Modal Speech Enhancement

Speech events involve the motions of several articulatory organs, such as the tongue, teeth, lips, jaw, and facial muscles, which can be leveraged for speech enhancement.

Audio and Visual

Researchers in (Michelsanti et al., 2021; Gao and Grauman, 2021) leverage the video from a camera in the environment to correlate the audio and visual for speech enhancement and separation. These approaches leverage deep learning and cross-modal embeddings to extract the target speech from noises. However, a visual sensor like a camera is not always available in daily usage.

Audio and Wireless

The authors in (Sun and Zhang, 2021; Zhang et al., 2021) use the smartphone's speaker to transmit an inaudible acoustic signal (i.e., higher than 17 kHz) and simultaneously receive the echo reflected by moving lips, which can be used to enhance the noisy audio. (Liu et al., 2021; Ozturk et al., 2021) use coupled mmWave and microphone recording for speech recognition rather than enhancing general speech contents. However, a mmWave radar is bulky and not common on HMWs, and the ultrasonic solution requires the user to fix the phone toward their mouth.

Audio and IMU

Some recent works have exploited the noiseless speech information from a bone conduction sensor or the accelerometer and the microphone recording for multi-modal speech enhancement.

(Wang et al., 2022) proposes a multi-modal DNN for speech enhancement trained by a self-collected dataset in Mandarin. However, the bandwidth of the vibration sensor is around 5 kHz, which is much higher than the one available on commercial devices (Sensortec, 2021). SEANet (Tagliasacchi et al., 2020) uses a DNN to augment acceleration from audio with the same transformation among all users. However, the diverse bone shapes can lead to different transformations and thus affect the generalizability. In addition, both works (Wang et al., 2022; Tagliasacchi et al., 2020) focus on the design of deep learning models and fail to evaluate the systems with real-world noises and unseen users.

3.2.3 Sensing Acoustic Signals

IMU or piezoelectric sensor (Maruri et al., 2018) can be installed on commercial devices to measure the contact sound, including unvoiced sound (Khanna et al., 2021), breathing sounds (Gupta et al., 2018), and eavesdropping smartphone/VR headset speaker (Ba et al., 2020; Wang et al., 2021; Su et al., 2021; Shi et al., 2021). Although various contact sensing technologies are developed for acoustic sensing, they can classify keywords and events only rather than reconstructing the full-spectrum audio. Previous works exploit remote acoustic sensing by correlating with several non-acoustic sensors (e.g., geophones, accelerometers, and gyroscopes) (Han et al., 2017), using the LiDAR on a vacuum robot to predict acoustic signals from the vibrations on the surrounding surfaces (Sami et al., 2020), or leveraging a mmWave Radar to achieve authentication (Li et al., 2020). However, they can only detect a given set of speech contents like digits and music. Measurements in (Anand and Saxena, 2018) show that smartphone's motion sensor can only capture air-conducted acoustic vibrations remotely in limited use cases.

3.3 Background

3.3.1 Audio Recording on HMWs

Although voice-related applications have been studied for decades, they usually exhibit unsatisfactory performance due to the following limitations. First, to save space and cost, most microphones installed on HMWs are omnidirectional instead of directional microphones due to their bulkiness and limitations in general use cases such as recording environmental sounds (e.g., Logitech 650e (Logitech, 2022), around 100 USD). The omnidirectional microphones pick up audio from any angle, so they inevitably record the interference signals, especially the speech of a competing speaker close to the target user. Second, although the latest HMWs usually equip multiple microphones for beamforming, the compact layout of the microphone array can impede the acoustic beamforming since the audio from different sources arrives at the microphones with only a minor time difference. Specifically, the microphone array's interval is much smaller than $\lambda_{min}/2$, while the λ_{min} is the minimum wavelength of the interested frequency band. Third, the microphones on HMWs are farther away from the user's mouth than wired earphones or standalone microphones since they are usually inserted into the ears or mounted around the eyes. Therefore, the received intensity of the speech audio can be significantly suppressed due to the slim air transmission. Lastly, the worldwide pandemic forces people to wear facial masks, which can significantly reduce speech volume by up to 7dB (Pörschmann et al., 2020). Hence, seeking another available transmission channel for robust speech sensing is desirable.

3.3.2 Bone-Conducted Vibrations

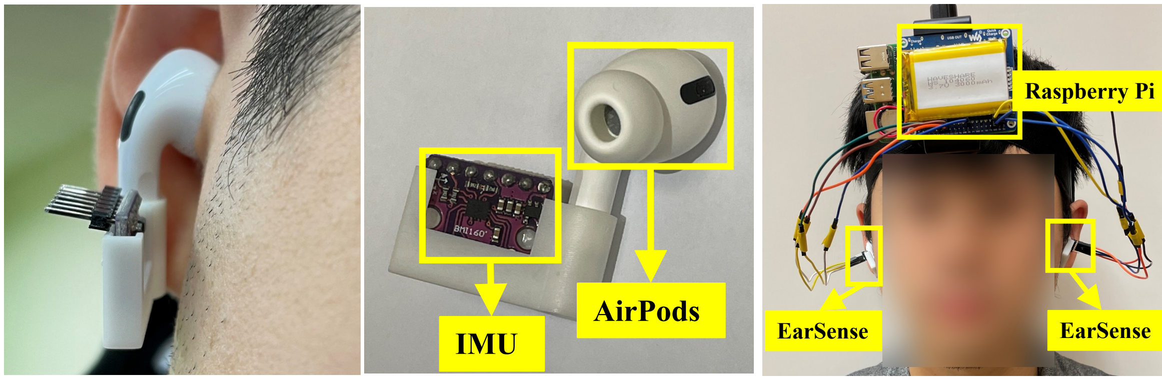
Existing commercial headphones leverage bone conduction (Wikipedia, 2022a) for inner ear hearing assistance. In this paper, we focus on the transmission from the vocal cord to the head skull, which can be detected by the contact microphone (Wikipedia, 2022b). The noise-less feature of bone conduction sound has been discovered (Tagliasacchi et al., 2020) with a bone conduction microphone. The professional bone-conducted microphone has also been applied in extremely harsh environments like battlefields or underwater applications (Ear, 2022; Bon, 2022). However, there are still challenges to leveraging it for speech enhancement on HMWs. Firstly, as depicted in (Chang et al., 2016; Won and Berger, 2005), the propagation through the head is complicated, resulting in an unstable response even for the exact location. Secondly, although a professional bone-conducted microphone can capture a clear voice at a high sample rate, they are too expensive (e.g., 60 US dollars (Ear, 2022; Bon, 2022)) and not available on commercial HMWs. The current commercial-level vibration sensor may not provide sufficient sample rate and precision, leading to frequency aliasing and a bad signal-noise ratio. Some approaches tackle the problems by duplicating the signals of low-frequencies to high-frequency domain (Dietz et al., 2002), utilizing the time stretching to generate high-frequency harmony (Nagel and Disch, 2009), or also trying to unfold the acceleration signals from low sample rate using deep learning (Wang et al., 2021). However, these approaches are far from achieving satisfying performance because the speech in high frequency contains extra information than that in low frequency. As a result, a simple recovery on the high-frequency part can amplify the noise of acceleration and degrade the quality of the generated audio.

3.4 Methodology

In this section, we first develop EarSense, an open-source wearable platform that has a similar setup to current commercial HMWs for data collection in Section. 3.4.1. Then, we exploit the bone-conducted vibrations and the audio signals recorded by EarSense under various settings in Section. 3.4.2. Finally, we overview the design of our proposed system in Section. 3.4.3, introduce Bone Conduction Function and multi-modal network in Section. 3.4.4 and Section. 3.4.5, respectively.

3.4.1 EarSense: An Open Source Data Collection Platform for HMWs

Most commercial HMWs do not provide APIs to collect raw acceleration data. Recently, Apple provided APIs (Apple, 2023) to collect motion data from AirPods earphones at 100 Hz, which is too low for speech recording and doesn't fully utilize commercial IMU. Hence, it is desirable to develop a new sensing platform that can: a) collect the acceleration and acoustic

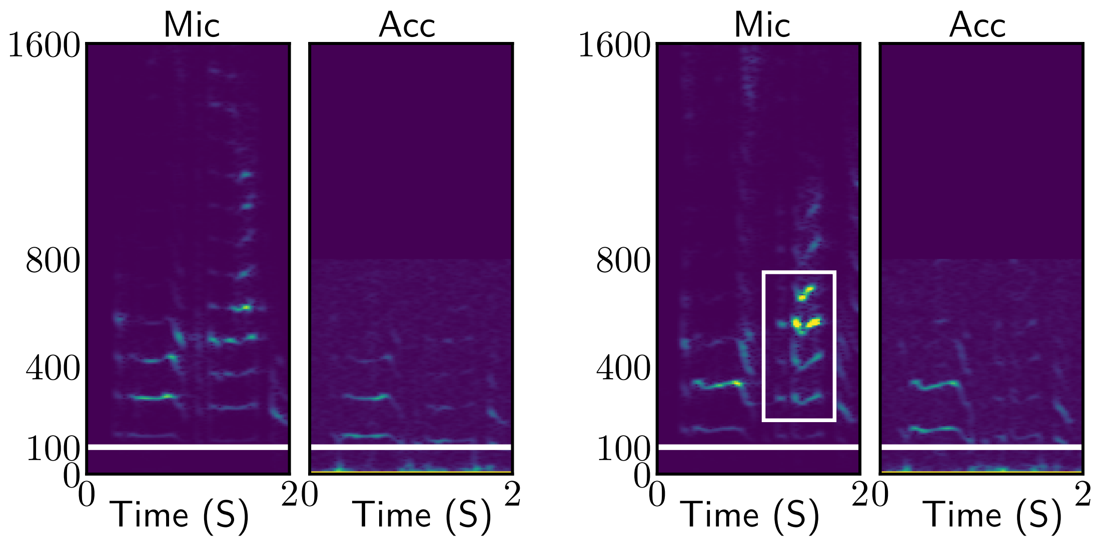


(a) EarSense *with an Airpods Pro.* **(b)** *Test Setup.*

Figure 3.2: EarSense *is an open-sourced data collector attachable to commercial HMWs for vibration sensing.*

data synchronously at a sample rate to recover speech information; b) have direct contact with the user's head like HMWs; c) have compatibility to test with existing commercial devices.

Fig. 3.2 shows the prototype of *EarSense* (Lixing et al., 2023), a new open-sourced sensing platform that equips a 3D-printed enclosure with an IMU sensor (i.e., Bosch BMI-160 (Sensortec, 2021)) inserted. EarSense has a flexible design to make it attachable to any commercial HMW device (e.g., AirPods Pro). EarSense captures the same vibration as the testing HMW and has no direct contact with the user's head (shown in Fig. 3.2a). We connect two EarSenses to a Raspberry Pi (RPi) with a battery for data collection. Fig. 3.2b shows a volunteer wearing the device on the head to avoid excessive vibrations caused by the connecting wires. A Python script on the RPi to control the EarSense(s) through the I2C protocol and store the data for offline processing. We only use the three-axis accelerometer provided by the IMU chip because the gyroscope and magnetometer are less related to the vibration. The default sample rate of the microphone and the accelerometer are 16 kHz and 1.6 kHz, respectively. Since commercial earphones like AirPods Pro doesn't support two-channel audio recording, EarSense records audio in a mono channel. We apply L2-Norm to the spectrograms of three-axis acceleration to extract the vibration intensity and avoid the impact of wearing position and user's motion.



(a) *The user is talking with no noise.* **(b)** *The user is talking with a competing speaker.*

Figure 3.3: *Impact of the competing speaker.*

3.4.2 Measurements of Bone-Conducted Vibrations

We measure the difference between audio and bone-conducted vibrations with EarSense under various settings, i.e., the noises caused by the competing speaker and user motion. In each experiment, we ask a volunteer to wear Apple AirPods Pro with EarSense attached (shown in Fig. 3.2b) in a meeting room with the size of 10 m^2.

Competing speaker. First, we ask the user to sit in the meeting room and speak while a loudspeaker is playing a pre-recording speech one meter from the user as the competing speaker. Note that the competing speaker is one of the most challenging environmental noises for HMWs since its spectrum is similar to the speech's, making it indistinguishable from the noise suppression algorithms. We set the volume of the competing speaker to be similar to the user, in which SNR is 3 dB. It is difficult to separate the two sources according to volume since the two speeches are mixed. Fig. 3.3a and Fig. 3.3b show the spectrograms of microphone audio and acceleration when the environment is quiet or contains a competing speaker when the user is talking, respectively. Note that we only show the spectrogram up to 1.6 kHz since there is slim energy in the frequency bands higher than

1.6 kHz of the microphone, and the accelerometer can only sense the signal with a frequency band of 800 Hz. The spectrograms show that acceleration has lower signal energy than microphone audio since acoustic vibration attenuates during bone conduction, especially for high-frequency vibrations. Compared with the silent environment, the audio spectrogram of microphone audio with a competing speaker shows strong distortions, as highlighted with the white box in Fig. 3.3b. However, the accelerometer spectrograms do not show this phenomenon because the competing speaker cannot produce bone-conducted vibrations. In summary, the accelerometer receives the user's voice through bone conduction only, excluding environmental sounds over the air, which is ideal for speech enhancement.

User motion. We also measure the noises caused by the user's motion. We ask the volunteer to walk while talking. Fig. 3.4 shows that walking causes low-frequency noises in the acceleration. The green boxes show the zoom-in result of the signals below 100 Hz. We can observe clear periodic fluctuations in low frequency (i.e., < 50 Hz), which are caused by the steps of the volunteer. In summary, the user's motion only affects the acceleration at lower frequencies, which does not affect our speech enhancement. In addition, we observe slight low-frequency fluctuation in acceleration in Fig. 3.3, although the volunteer is asked to stand still in this experiment. Since most of the human speech is above 85 Hz (Wikipedia, 2020b), the removal of audio in low frequency (i.e., the white horizontal lines in Fig. 3.3) can effectively reduce the interference caused by human motion.

Frequency Response. We further explore the frequency response of bone-conducted vibrations among users. Specifically, we measure the frequency response of bone-conducted vibrations as follows: First, we compute the spectrums of audio and vibration data, which is denoted by S_{Mic} and S_{Acc}, respectively. Then, we compute the Bone Conduction Function denoted by the frequency response of bone-conducted vibrations by S_{Acc}/S_{Mic} at every frequency band. Fig. 3.5 shows two volunteers' frequency responses, in which the error bars represent the averages and variances. The acceleration shows higher sensitivity than the microphone within the frequency of 200Hz $\sim$ 500Hz, while lower sensitivity at a higher frequency ($>$ 500Hz) due to the absorption by the head skull. In addition, the frequency

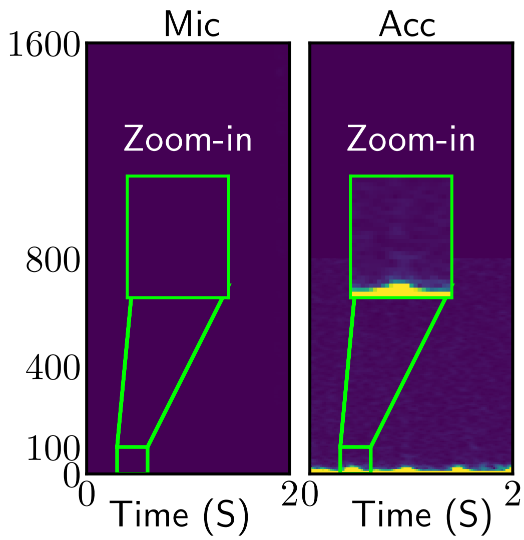


Figure 3.4: *Spectrograms of microphone and acceleration when the user is walking.*

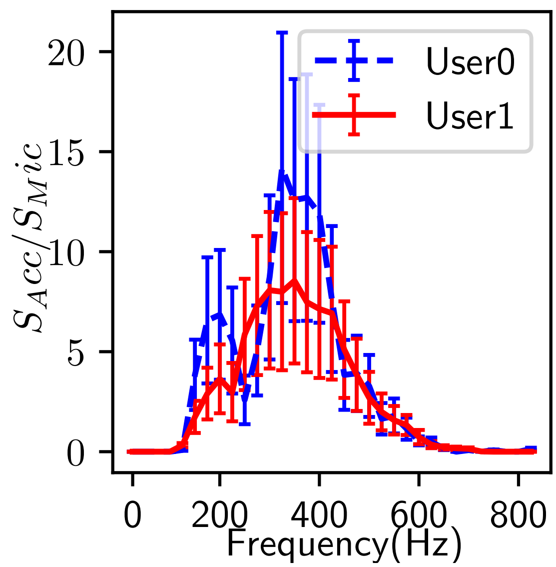


Figure 3.5: *Frequency responses of two users' bone-conducted vibrations.*

response keeps significant similarity among users, indicating the generalization to other users. Hence, the Bone Conduction Function should consider inter-people diversity and inter-people similarity.

Locations on the head. Bone-conducted vibration exists at various locations on the head. As shown in Fig. 3.6, we select ten unique locations of interest on the head and measure the intensities of received bone-conducted vibrations. In particular, the nine locations are #1 *upper ear*, #2 *eyebrow*, #3 *cheekbones*, #4 *ear*, #5 *temporomandibular joint*, #6 *cheek*, #7 *temple*, #8 *back of the head*, and #9 *nose*. In addition, we tape EarSense on the interior of the pad of an over-ear headphone, which is denoted as #10. Among those positions, some are compatible with commercial HMWs like glasses, headphones, and VR headsets. Fig. 3.7 illustrates the corresponding positions on HMWs. The numbers on the HMWs correspond to the locations in Fig. 3.6. We attach EarSense to the HMWs (when applicable) or on the face for all the following tests. We compute Pearson correlation coefficients between the audio and the acceleration on each location and mark the values using a color map in Fig. 3.6. We observe that bone-conducted vibrations exist at most locations of the head. In particular, locations closer to the mouth show higher correlations, indicating a larger possibility of extracting

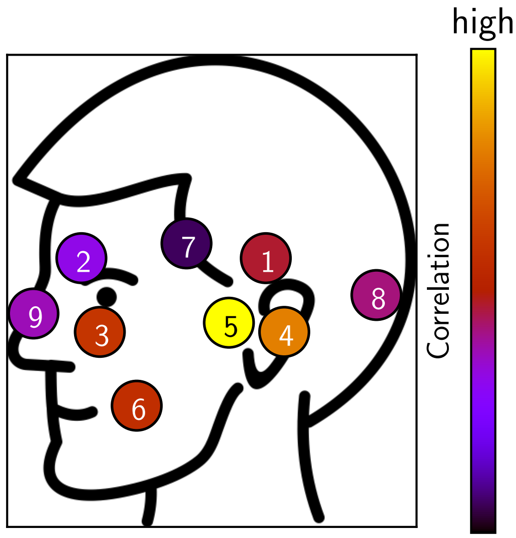


Figure 3.6: *Intensities of received bone-conducted vibrations on ten locations of the head.*

Figure 3.7: *Potential placements on devices.*

clean speech.

In summary, our measurements show that bone-conducted vibration has the following characteristics: a) Bone-conducted vibration is robust to environmental voice and only captures the user's speech; b) The user's motion only generates vibrations lower than 85 Hz; c) Bone-conducted vibration has suppressed low and high-frequency audio, which varies across users and within the same user; and d) We can receive audio from multiple locations on the head.

3.4.3 Overview of VibVoice

Motivated by our findings in Section. 3.4.2, bone-conducted vibration is a promising complementary sensing modality to microphone recording for speech enhancement under environmental noises. However, the limited sample rate of HMW's accelerometer and diverse frequency responses of bone-conducted vibration introduce challenges for multi-modal speech enhancement. On the other hand, although it is possible to adapt existing audio-only DNN-based speech enhancement models (Michelsanti et al., 2021; Sun and Zhang, 2021; Zhang et al., 2021; Liu et al., 2021; Ozturk et al., 2021) to take both inputs, there

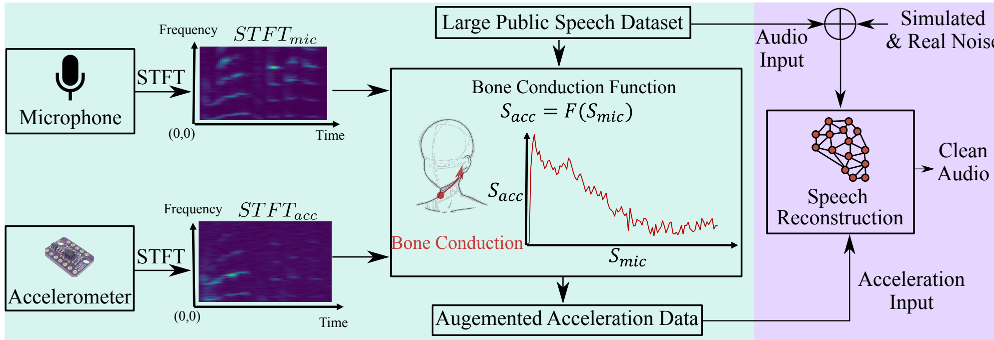


Figure 3.8: *The overview of VibVoice system. We first augment the acceleration data using a large public speech dataset with estimated Bone Conduction Functions (e.g., the greed box). Then, we train a multi-modal DNN for speech reconstruction (e.g., the purple box).*

is no large dataset with paired acceleration and audio on HMWs available.

We design *VibVoice*, a speech enhancement system for HMWs that leverages both the microphone recording and the speech information from the bone-conducted vibrations. Fig. 3.8 overviews the design of VibVoice, which contains three components: estimation of Bone Conduction Function, data augmentation, and speech reconstruction network. First, we estimate a set of Bone Conduction Functions using a small paired acceleration-audio dataset collected from volunteers in Section. 3.4.4. We note that this estimation does not resort to black-box deep learning approaches, which require a huge amount of training data. Instead, we take advantage of prior knowledge that a frequency response exists between the acceleration and audio spectrograms. Then, we augment the acceleration data using the public audio dataset *LibriSpeech* (Panayotov et al., 2015) with Bone Conduction Functions in Section. 3.4.4. Finally, after generating a large amount of paired acceleration-audio dataset, we train a DNN model which can reconstruct the clean audio from microphone audio and the acceleration data in Section. 3.4.5.

3.4.4 Bone Conduction Function

Function Formulation

Measurements in Section. 3.4.2 show that the acceleration contains acoustic vibrations caused by bone conduction effects. There are two characteristics of acoustic vibrations: a limited sample rate up to 800 Hz and diverse amplitude suppression levels at frequency bands. We model the vibration sensed by the accelerometer as follows:

$$s_{acc} = f(s_{mic}) + \epsilon_{acc} = f(s_{speech} + \epsilon_{mic}) + \epsilon_{acc}, \tag{3.1}$$

where s_{acc} and s_{mic} are the raw data captured by the accelerometer and the microphone, respectively; s_{speech} denotes the ground-truth (clean) speech audio; ϵ_{acc} and ϵ_{mic} are environmental noises captured by the accelerometer and the microphone, respectively; and f is the Bone Conduction Function.

Function Estimation

To estimate the Bone Conduction Function, we recruit volunteers and record a five-minute speech for each person with EarSense and AirPods Pro. The volunteers are asked to read from a daily conversation material (USA, 2022) in a silent environment. We split paired data into 5-second windows, and each window can contribute one Bone Conduction Function.

First, we compute the spectrograms $STFT_{acc}$ and $STFT_{mic}$ of s_{acc} and s_{mic} for each window using short-time Fourier transform (STFT), respectively. Then, we apply Otsu's method (Otsu, 1979) to $STFT_{acc}$ and $STFT_{mic}$ for automatic image thresholding, and then discard the bins whose value is lower than a threshold to remove outlier noises. We note that our thresholding setting can leverage temporal information, which can better diminish the noise than a constant threshold on the frequency or time domains. Furthermore, we perform the pixel-wise division between $STFT_{acc}$ and $STFT_{mic}$ to obtain a response spectrogram, selecting the lower frequency part of the audio spectrogram to align to the acceleration spectrogram. We model the Bone Conduction Function using the Gaussian distribution in the frequency domain since the frequency response (as shown in Fig. 3.5) has a non-trivial variance due to the complex structure of the head skeleton (Chang et al., 2016). In particular, we compute $f = S_{acc}/S_{mic}$ for the corresponding time window in $STFT_{acc}$ and $STFT_{mic}$, respectively. The function $f \sim N(\mu, \sigma^2)$, in which μ and variance σ contribute to the contour and fluctuation of frequency response, respectively. We measure μ and σ for each time

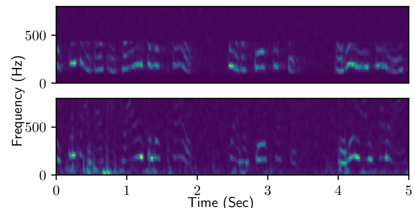


Figure 3.9: *Augmented (upper) and real (lower) Acceleration.*

window to construct the parameter pool of Bone Conduction Functions.

Data Augmentation with Bone Conduction Functions

We develop a data augmentation approach with Bone Conduction Functions described in Section. 3.4.4. Note that we cannot apply the inverse Bone Conduction Function to turn the acceleration back to audio due to the significantly limited sample rate of acceleration data. In addition, the frequency band (larger than 500 Hz) has very slim energy, which can cause unreasonable large energy after the inversion. We utilize these functions to generate acceleration signals using a large-scale audio dataset, i.e., LibriSpeech (Panayotov et al., 2015). To be specific, for each audio clip, we first select a Bone Conduction Function (i.e., a list of means and variances over different frequencies) from the pool randomly. Then, we restore the frequency response from the Gaussian distribution of given parameters. Lastly, we can augment the audio to synthetic acceleration data by directly multiplying the frequency response.

Fig. 3.9 shows the spectrograms of augmented acceleration and real acceleration signals, respectively. The augmented spectrogram is close to the real acceleration spectrogram. We compute the similarity by calculating the mean of the absolute distance of all pixels in the whole spectrogram and divide it by the largest value of the real acceleration spectrogram. The average error for all volunteers is only 4.5%, which indicates that our proposed acceleration augmentation is reliable.

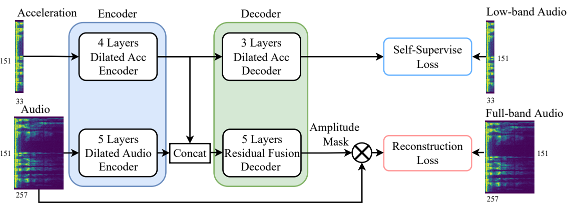


Figure 3.10: *Overview of our proposed multi-modal network for speech enhancement.*

3.4.5 Multi-Modal Speech Reconstruction

Since the sample rate of the microphone is 16 kHz while the accelerometer is only 1.6 kHz. The reconstruction of high-quality audio from acceleration is an ill-posed task since it requires predicting the lost patterns of the high-frequency band. Moreover, as we have already generated a large-scale paired acceleration-audio dataset, we can unleash the strong feature extraction capability of DNNs for training. Compared with hand-crafted signal processing approaches, deep learning is robust to diverse inputs. Specifically, we adapt the multi-modal fusion paradigm (Sun and Zhang, 2021; Zhang et al., 2021) and encoder-decoder architecture based on U-net (Ronneberger et al., 2015) to build the deep learning model. Fig. 3.10 overviews of the DNN design, which is described in detail as follows.

	Encoder				Decoder		
Layer	1	2	3	4	1	2	3
Filters	16	32	64	128	64	32	16
Kernel	3	3	3	3	3	3	3
Scale	1	1	0.5	0.5	1	2	2

(a) *Acceleration branch.*

	Encoder					Decoder				
Layer	1	2	3	4	5	1	2	3	4	5
Filters	16	32	64	128	256	128	64	32	16	1
Kernel	3	5	5	5	5	5	5	5	5	3
Scale	0.5	0.5	0.5	0.5	0.5	2	2	2	2	2

(b) *Audio branch.*

Table 3.1: *The design of DNN for multi-modal speech reconstruction.*

Architecture

Table 3.1a and 3.1b show the hyperparameters of our multi-modal neural network of the acceleration and audio branches, respectively. The details are illustrated as follows.

Encoder. We use two convolutional networks (CNNs) to encode the high-level features from the two modalities, respectively. Then, we concatenate the features from two encoders along the channel dimension. We stack basic blocks to construct the encoder, where each block consists of a 2D convolutional layer, batch normalization, ReLU activation, and max-pooling. We also design a residual shortcut of the block input before the last deconvolution layer to expedite the training. The filters increase when the layer gets deeper to extract more diverse features. To increase the reception field to capture the harmonic pattern of the whole spectrogram, we use dilated convolution rather than the conventional one. Since the sample rate of audio data (i.e., 16 kHz) is ten times higher than that of the acceleration data (1.6 kHz), the size of the audio spectrogram on the frequency axis is also ten times larger. We note that the feature maps from the two modalities should have the same shape at the end of the encoders. Therefore, we discard the parts of the audio spectrogram whose frequency is higher than 6.4 kHz, making the remaining audio frequency band exactly eight times larger than the acceleration data (i.e., 0.8 kHz). In addition, the audio subnet (i.e., encoder) has three more down-sample operations than the acceleration branch to make the output sizes of the two modalities' features consistent.

Decoder. We design two decoders: a fusion decoder and an auxiliary decoder. The decoder has the same design of blocks as the encoder, but the stacked blocks have a decreasing number of filters but increasing sizes of the output tensor. The fusion decoder receives both modalities' features via a concatenation and outputs a spectrogram mask. To obtain the constructed clean spectrogram, this mask will be overlaid on the original noisy audio spectrograms by element-wise multiplication. The auxiliary decoder only receives the feature of the accelerometer and predicts the low-frequency part of clean audio. The self-supervised loss can prevent the acceleration encoder from being dominated by the audio input, given that the noisy input audio is similar to the clean audio to acceleration by nature (i.e., both are audio).

Waveform reconstruction

After obtaining the enhanced spectrogram from the multi-modal decoder, the last step is converting the spectrogram to the waveform using inverse short-time Fourier transform (ISTFT). The DNN of VibVoice only reconstructs the magnitude of the spectrogram. Since the phase of the speech spectrogram is hardly predictable, we use the same phase of the noisy audio signal for the output. Our insight is that although the audio phase can be polluted in a noisy environment, our reconstructed magnitude can attenuate the unwanted time-frequency pixels in spectrograms, which restricts the interference of the noisy phase. Although several works (Sun and Zhang, 2021; Zhang et al., 2021) try to estimate the clean phase by a subnet or a pre-defined algorithm, our tests show that the predicted phase is unstable, and their improvements are marginal and computationally expensive.

Model Training

We generate a pool of Bone Conduction Functions by performing cubic interpolation on existing functions from our dataset. We use Gaussian distribution to alleviate our model's overfitting and improve the generalizability among users. In addition, we augment the data with Gaussian distribution based on each frequency band to preserve the low-pass features. Thus, the limited Bone Conduction Functions can augment a large acceleration-audio dataset from an existing public dataset with diverse features. This dataset can extract the basic features of multiple modalities, but it is not enough for deployment because of the difference between the target user and our pool of Bone Conduction Functions. Therefore, we collected another acceleration-audio paired dataset from a different group of volunteers to train the model, which is also used to evaluate VibVoice's performance. Note that VibVoice does not require any data from the target user either during the Bond Conduction Function estimation or model training. All evaluations adopt leave-one-out validation. We pick one user as the testing dataset, while the others are regarded as the training dataset.

Although the training of VibVoice does not require any data from the target user, one-shot data from the target domain can further improve the overall performance. This data collection can share the recorded data during the setup of HMWs, e.g., the personalized model for calling voice assistants, which incurs no extra overhead to the user. In addition, VibVoice can record data when the user is speaking in a quiet environment. As the data collection doesn't require specific content, our system can unobtrusively record user-specific data during the long-time usage of HMWs and improve the performance of the target user continuously.

The fusion decoder uses the STFT Loss (Defossez et al., 2020). We use y and $\hat{y}$ to represent the clean and enhanced signals. The STFT loss can be denoted as follows:

$$\begin{aligned} L_{stft}(y,\hat{y}) &= L_{sc}(y,\hat{y}) + L_{mag}(y,\hat{y}), \\ L_{sc}(y,\hat{y}) &= \frac{||STFT(y)| - |STFT(\hat{y})||_F}{||STFT(y)||_F}, \\ L_{mag}(y,\hat{y}) &= \frac{1}{T}|log|STFT(y)| - log|STFT(\hat{y})||_1, \end{aligned} \tag{3.2}$$

where L_{sc} refers to convergence (sc) loss and L_{mag} refers to the magnitude (mag) loss. We use Mean Square Error (MSE) as the training loss for the auxiliary decoder. The fusion decoder and auxiliary decoder targets are full-band clean spectrogram and lower-band clean spectrogram, respectively. The final loss is formulated as follows:

$$L(y,\hat{y}) = L_{stft}(y,\hat{y}) + |y_{lowband} - \hat{y}_{lowband}|_2 \times \lambda, \tag{3.3}$$

which is the joint loss function that covers both fusion and auxiliary decoder. We set λ to 0.05 to balance the scales of two losses.

3.5 Evaluation

3.5.1 Experiment Setup

We use EarSense introduced in Section. 3.4.1 to collect audio and acceleration signals simultaneously[1]. The volunteers are asked to wear EarSense and speak in a meeting room with the size of 10 m^2. The reading material is selected from daily English conversations (USA, 2022). Each volunteer reads the material for 30 seconds and repeats it 20 times, generating ten minutes of data. For data augmentation, we recruit eight volunteers to generate a pool of Bone Conduction Functions and use it to augment 100 hours of data from LibriSpeech (Panayotov et al., 2015). In addition, we recruit another 15 volunteers for training and testing. For each experiment, we adopt the leave-one-out validation, i.e., training the model for each user with the dataset except that user. The volume of our collected speech is between 60dB to 70dB, which is the same as regular conversations (Wha, 2022). To test VibVoice's robustness to diverse noises, we use three categories of noises with balanced possibility, including environmental noises (50 classes) (Piczak, 2022), competing speakers from another subset of LibriSpeech (Panayotov et al., 2015), and 20 songs with languages of English, Mandarin, and Japanese. We apply a random room impulse response from dataset (Ko et al., 2017), containing point-source noises, real isotropic noises, and simulated the noises of 600 rooms. We implement VibVoice using PyTorch and train it on a PC with an Intel i9-12900K CPU and two Nvidia RTX 3090 GPUs. The model is trained with a step learning rate scheduler with a learning rate of 0.001 and an Adam optimizer. The epoch number is 30. The length of the STFT window and the overlap for the audio are 640 and 320, while 64 and 32 are for the acceleration. We open-source the implementation and data in (Lixing et al., 2023).

3.5.2 Baseline

We deploy two baselines, i.e., FullSubNet (FSN) (Hao et al., 2021) and SEANet (SN) (Tagliasacchi et al., 2020). FSN and SN are two state-of-the-art speech enhancement ap-

[1]The experiments that involve human subjects have been approved by the IRB of the authors' institution (CUHK-SBRE-21-0570).

proaches using audio-only and audio-acceleration inputs, respectively. We train SN using our dataset since the one used in its paper is not available to the public. Specifically, we train SN using acceleration data with a sample rate of 1.6kHz to ensure it is the same as VibVoice. Note that SN indicates that it supports acceleration data with a wide range of sample rates.

3.5.3 Evaluation Metric

We use three evaluation metrics, i.e., Perceptual Evaluation of Speech Quality (PESQ), Signal Noise Ratio (SNR), and Log-Spectral Distance (LSD), which are described as follows:
Perceptual Evaluation of Speech Quality (PESQ) is a popular test standard for automatically assessing the user experience of speech quality, defined in P.862 standard by International Telecommunication Union (Rix et al., 2001). The results generated by PESQ represent the opinion scores that range from 1 (bad) to 5 (excellent). We use the wide-band version to evaluate the full-band speech quality in the evaluations.
Signal Noise Ratio (SNR) is a metric widely used in signal processing that can be measured as follows: $SNR(x,y) = 10 * log_{10}(\frac{y}{x-y})^2$, where x and y denote the estimated and clean audio, respectively. SNR compares the desired signal and the deviation. A higher SNR value means better audio quality. We use the scale-invariant SNR in the implementation to reduce the impact of scale.
Log-Spectral Distance (LSD) measures the quality of frequencies between the reconstructed audio and the ground truth audio with: $LSD(x,y) = \frac{1}{L}\sum_{l=1}^{L}\sqrt{\frac{1}{K}\sum_{k=1}^{K}(X(l,k) - \hat{X}(l,k))^2}$, where l and k denote the time and frequency index of the spectrogram, respectively; $X = log(|STFT(y)|^2)$, and $\hat{X} = log(|STFT(x)|^2)$. A higher LSD value means lower audio quality.

3.5.4 Overall Performance

We investigate VibVoice's performance by mixing clean speech with noise audio randomly picked from the noise dataset. We set the SNR of the noisy data ranges from 0dB to 20dB, with an average of 10dB, which covers a wide spectrum of noise levels.

Target user calibration. VibVoice can work out of the box, while data from the target user can further improve performance. Fig. 3.11 shows the performance of VibVoice and the two baselines with different amounts of data from the target user, in which zero means no target-user data is used during the training. The results show that longer target-user data can improve performance for all approaches, and VibVoice performs better than baselines with the same amount of calibration data. The calibration data can be collected continuously during usage without the user's inputs/operations (c.f., Section 3.4.5). VibVoice achieves the best perceptive performance according to PESQ and the best-reconstructed spectrogram according to LSD. VibVoice and FSN have similar SNR results since they output similar signals in the time domain compared to the clean one.

Noise type. We evaluate the impact of different types of noises in Fig. 3.12, i.e., environmental noises, competing speakers, and music. The result shows that VibVoice performs better under all noises and metrics except for the SNR with music noise. This is because the complex and dynamic spectrum of the user's speech and music's vocals introduce minor fluctuations in high frequencies, reflecting large fluctuations in SNR.

Noise level. We test VibVoice's performance under noise levels of low (10 dB), medium (5 dB), and high (0 dB with only speech noise). The results in Fig. 3.13 show that VibVoice has better performance improvements, especially when the noise is challenging, i.e., 21% improvement on PESQ and 26% improvement on SNR. This is because the acceleration can more robustly identify the target speech, whereas the audio-only solution is difficult to differentiate sound with a similar pattern (e.g., strong speech noise).

Noise source. We evaluate the impact of different numbers of noise sources by repeatedly mixing clean audio with random audio clips. Fig. 3.14 shows that VibVoice's performance degrades as the number of noise sources increases but still outperforms all the baselines.

Temporal stability. We further examine how VibVoice performs for the same user over time. Note that the offset of sensor placement and minor changes in speech can cause a slight change. We collect ten-minute data from three volunteers twice, six months apart.

The results show that the performance of VibVoice has negligible changes, from 2.6 to 2.5 for PESQ, 15.7 to 15.5 for SNR, and 4.3 to 4.6 for LSD. Besides, VibVoice outperforms FSN, whose performance is 2.1 for PESQ, 15.2 for SNR, and 11 for LSD. The results affirm that VibVoice is robust to temporal changes.

Airway blockage. The blockage of air transmission caused by personal protective equipment like facial masks can significantly suppress the user's speech volume. We ask three volunteers to test VibVoice by speaking the same content with and without facial masks. The result shows that VibVoice's performance only degrades by 0.05 ($< 2\%$) for PESQ, 0.4 ($< 3\%$) for SNR, and 0.1 ($< 3\%$) for LSD when the subject wears the mask. VibVoice gets a more significant margin than FSN, whose performance is 2.15 for PESQ, 13.8 for SNR, and 10 for LSD. This aligns with our expectation since VibVoice uses bone-conducted vibration that is not affected by air transmission.

Variances among users. Speech and bone-conducted vibration can differ across users due to vocal features, head skull shapes, body fat, etc. Fig. 3.16 shows the performance of VibVoice across 15 users. VibVoice shows stable and significantly better performance in PESQ compared to baseline and comparable performance in SNR.

Sensing location. We test VibVoice when EarSense is placed in ten locations on the head as defined in Fig. 3.6, validating VibVoice's effectiveness for different HMW devices. The bars in Fig. 3.15 show VibVoice's performance at each location. The red line represents the performance of the baseline at #4 *ear*. The results show that VibVoice achieves satisfactory performance at all locations in PESQ. Note that locations like #1 *upper ear*, #2 *eyebrow*, #5 *temporomandibular joint*, #7 *temple*, and #10 *interior of the pad of the headphone* show similar or slightly lower SNR than the baseline, which is because the vibration intensity is slim due to their far distance to the audio source.

Data augmentation effectiveness. We evaluate how VibVoice's data augmentation reduces the amount of paired training data needed. We compare the performance of training VibVoice from a) paired data of 18 to 180 hours and b) data augmented from three hours of paired data. The dataset (Wang et al., 2022) is a large-scale Chinese acceleration and

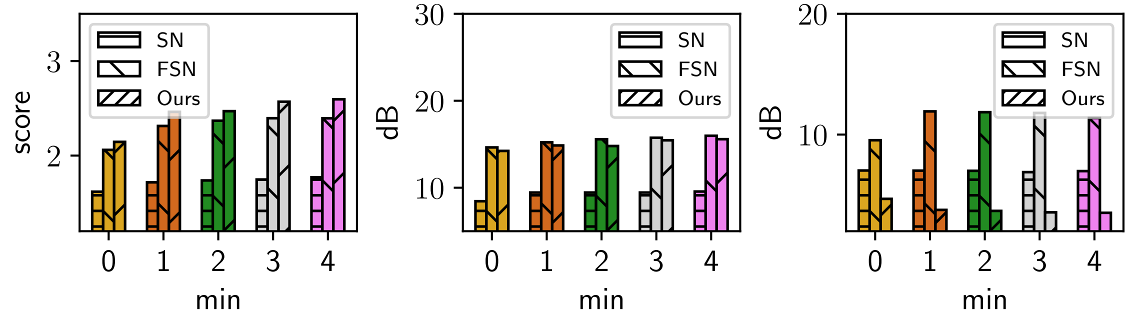
	PESQ	SNR	LSD
VibVoice	2.6	15.6	3.5
w/o auxiliary decoder	2.5	15.1	4.4
w/o augmentation	1.9	14	5
w/o Gaussian approx	2.4	15.2	4.2
Accelerometer sample rate: 1200 Hz	2.47	14.4	4.2
Accelerometer sample rate: 800 Hz	2.45	14.3	4.5
Accelerometer sample rate: 400 Hz	2.4	14.2	4.4

Table 3.2: *Ablation study.*

audio dataset collected by earphones, and the bandwidth of acceleration is around 5kHz. The results in Fig. 3.17 show that VibVoice with data augmentation can achieve better performance with $\sim 24\times$ less paired data.

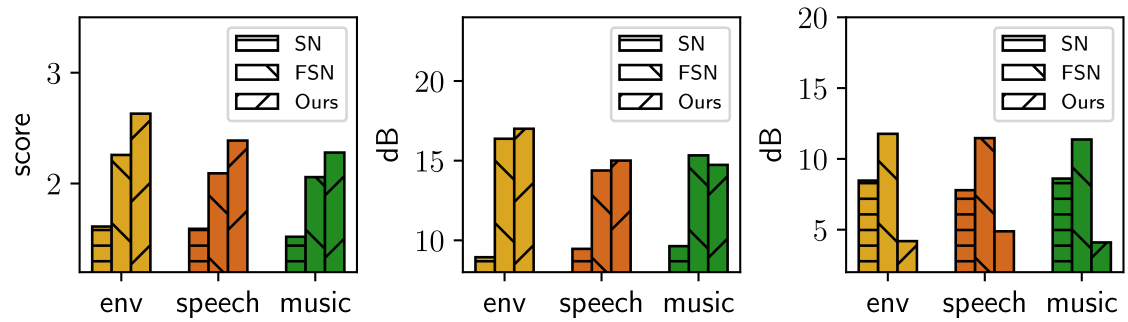
Summary. VibVoice outperforms FSN and SN by up to 21% for PESQ when the noise volume is low, where the PESQ for VibVoice and FSN are 2.7 and 2.21, respectively. VibVoice outperforms the baselines up to 26% for SNR when the noise is speech with high volume, where the SNR of VibVoice and FSN are 2.0 and 1.6, respectively. In addition, VibVoice outperforms the baselines 50$\sim$80% in LSD under most impact factors, indicating the efficiency of our multi-modal design and novel data augmentation. VibVoice has slightly lower performance in SNR for some cases as it can be biased due to the similar spectrum of the user's speech and music's vocal. The dynamic also introduces minor fluctuations. In comparison, LSD evaluates the whole band without preference, so VibVoice outperforms the two baselines by a large margin. We further evaluate the perception of real users through a user study in Section. 3.6.

Compared with the strong baseline FSN, VibVoice achieves good performance in various cases. Compared with multi-modal baseline SN, whose network design has limited capability to recover speech information as VibVoice outperforms it by a large margin. Besides, as discussed in Section. 3.2.2, the data augmentation of SN does not consider the user-specific bone conduction effects, while VibVoice combines the knowledge from extensive measurements and individual features from the target user.



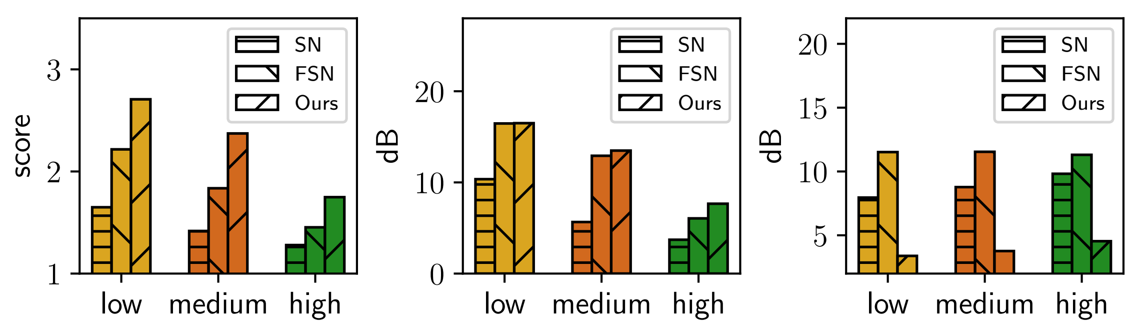
(a) *PESQ.* **(b)** *SNR.* **(c)** *LSD.*

Figure 3.11: *Imapct of calibration time.*



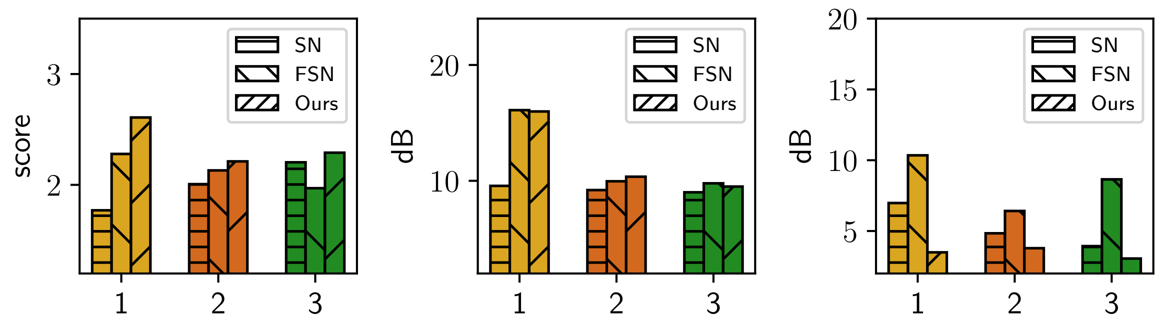
(a) *PESQ.* **(b)** *SNR.* **(c)** *LSD.*

Figure 3.12: *Impact of noise types.*



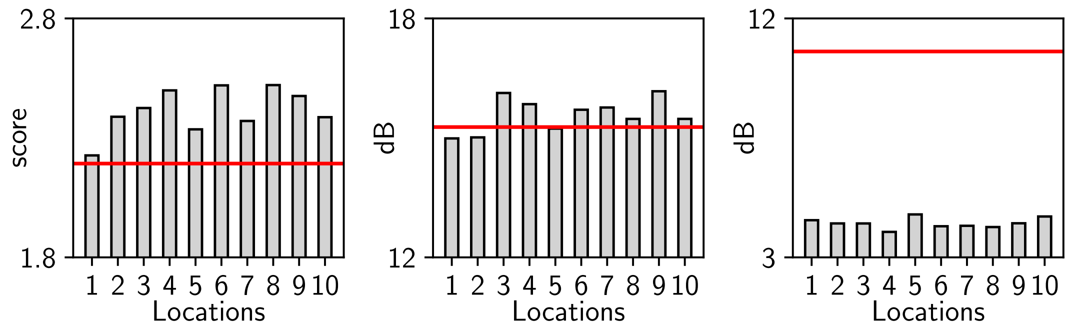
(a) *PESQ.* **(b)** *SNR.* **(c)** *LSD.*

Figure 3.13: *Impact of noise levels.*



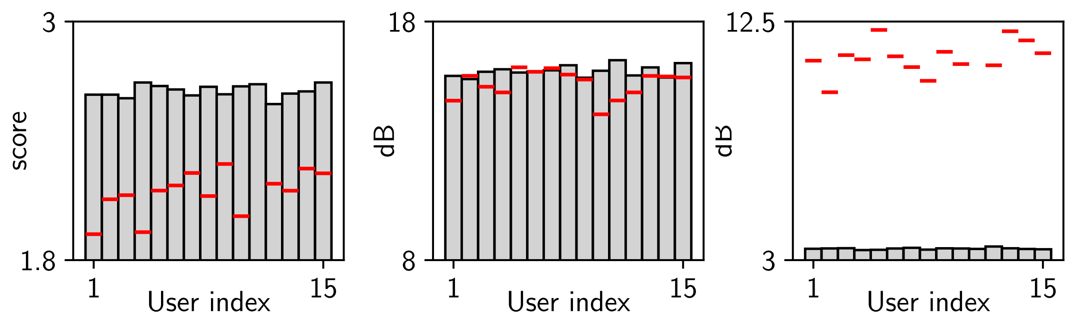
(a) *PESQ.* **(b)** *SNR.* **(c)** *LSD.*

Figure 3.14: *Number of noise sources.*



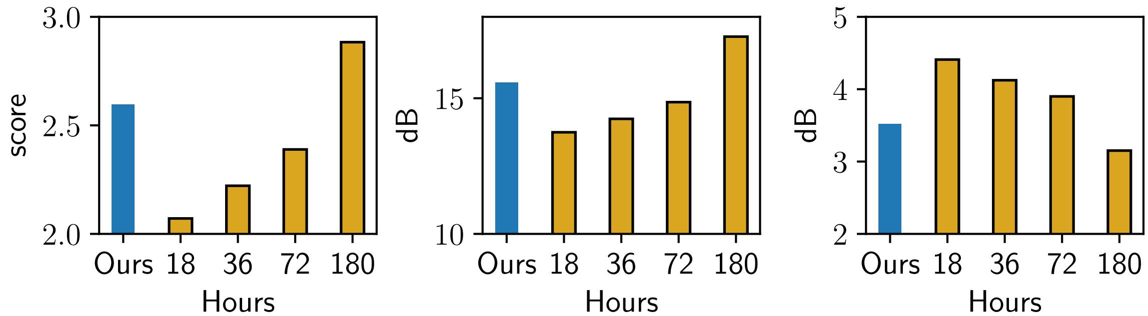
(a) *PESQ.* **(b)** *SNR.* **(c)** *LSD.*

Figure 3.15: *VibVoice on different head locations. Red line: the performance of the baseline, i.e., FullSubNet.*



(a) *PESQ.* **(b)** *SNR.* **(c)** *LSD.*

Figure 3.16: *VibVoice on different users. Red line: the performance of the baseline, i.e., FullSubNet.*



(a) *PESQ.* **(b)** *SNR.* **(c)** *LSD.*

Figure 3.17: *Effectiveness of data augmentation. Blue bars: VibVoice using less than three-hour paired data with augmentation. Yellow bars: VibVoice using 18- to 180-hour paired data without augmentation.*

3.5.5 Ablation Study

We conduct an ablation study to understand the performance of different design components in VibVoice. The performance of VibVoice without different components is listed in Table 3.2.

No auxiliary decoder. First, we remove the self-supervise loss, meaning the audio may dominate the model. The results indicate that the variant slightly degrades by 0.1 in PESQ, 0.3 in SNR, and 0.8 in LSD, respectively.

No data augmentation. Second, we remove the data augmentation based on Bone Conduction Function. The performance significantly degrades to 1.9 in PESQ, 14 in SNR, and 5 in LSD. According to the definition of ITU and mean opinion score (MOS), the audio quality is poor when the score drops from 2.5 to 1.9 (Wikipedia, 2020a). This result aligns with our motivation that a small-scale self-collected dataset is insufficient to train a strong neural network.

No Gaussian approximation. Third, the Bone Conduction Function is modeled by only the mean, while the variance is zero. The performance drops 0.2 in PESQ, 0.4 in SNR, and 0.5 for LSD, indicating that our Gaussian approximation is close to the nature of the Bone Conduction Function.

Lower sample rate. The frequency response in Fig. 3.5 shows that the speech information from above 600 Hz is very limited. Hence, we evaluate the performance of VibVoice with a lower sample rate by downsampling the acceleration data to 1200 Hz, 800 Hz, and 400 Hz. The results show that VibVoice is robust to various sample rates, which consume less power in processing and communication. When the sample rate is 800 Hz, the output's PESQ only degrades 10%.

3.5.6 On-Site Evaluation

In the following, we conduct extensive experiments to validate the performance of VibVoice with ongoing noise in the wild. The volunteers are asked to read the same content as the experiments in Section. 3.5.4. Apart from the environmental noises, we use a smartphone (i.e., Huawei P30) to play recorded speech and ask a volunteer to speak one meter away as the interference. The speech content of the competing speaker is irrelevant to the target speech. We collect 10 minutes of data with about 3dB SNR at each test location. Unlike synthetic noisy speech, we can neither capture the ground truth of clean speech and evaluate the metrics like SNR and PESQ nor train the model. Instead, we use the result of Automatic Speech Recognition (Ravanelli et al., 2021) to evaluate the quality of the speech. We use Word Error Rate (WER $= \frac{S+D+I}{N}$) as the evaluation metric, where S is the number of substitutions, D is the number of deletions, I is the number of insertions, and N is the number of words in the reference. The higher the value of WER, the lower the audio quality. To better evaluate the performance of VibVoice, we further define the relative improvement of WER in percentage as $\frac{(WER_o - WER_v)}{WER_o} \times 100\%$, where WER_o refers to the WER of the original audio, WER_v refers to the WER of VibVoice.

Environments

We evaluate VibVoice in diverse indoor and outdoor environments, including the meeting room, corridor, stairs, lumber room, and railway station. We note that the experiment conducted in the real world naturally contains diverse environmental noises (e.g., train, car, construction, wind) and competing speakers. The results in Fig. 3.18 show that VibVoice

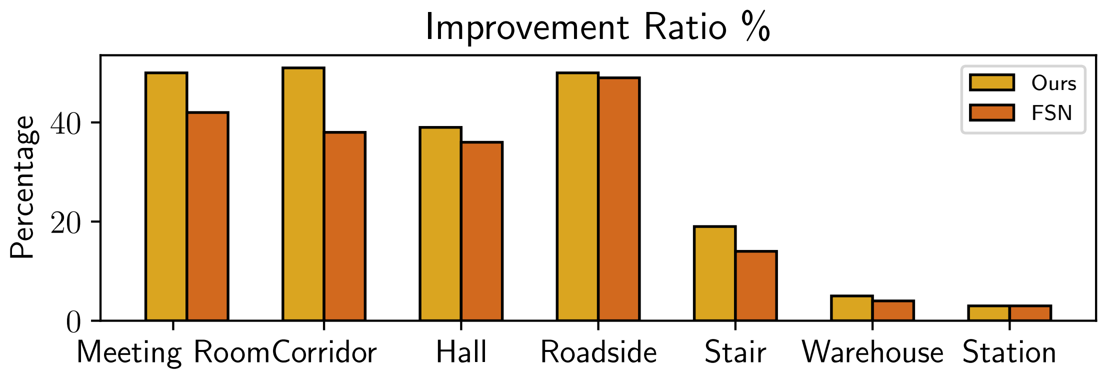


Figure 3.18: *Different environments.*

effectively improves speech quality in most environments. Most existing works have good performance in statistical noises like machinery noises, however, they perform badly when there are competing speakers in the same room, which is a common indoor scenario of voice communication. Although speech enhancement is challenging in small spaces like a meeting room and corridor due to the strong multi-path effect, VibVoice can still improve the speech quality and achieve an improvement ratio of over 50%, and 48%, respectively. VibVoice also achieves a 37% improvement for the hall, 29% for the outdoor square, and 19% in the scene of the outdoor stair when there is intermittent construction noise. We note that in both the warehouse and station, the improvement of VibVoice is marginal. This is mainly because that AirPods already perform well in these scenarios. In comparison, VibVoice keeps outperforming FSN with much less latency.

Earphones

We deploy VibVoice on three popular TWS earphones, i.e., Apple AirPods Pro, Huawei FreeBuds Pro, and Samsung GalaxyBuds Pro. During the experiment, we turn on all available acoustic signal processing algorithms provided by the manufacturers, including speech enhancement and noise suppression. The results show that the WER with a competing speaker is 60% for AirPods, 66% for FreeBuds, and 160% for GalaxyBuds, respectively. Due to the different designs of the earphones, we do not implement VibVoice on other earphones.

	VibVoice	FSN	SN
Desktop CPU	0.05	0.27	0.5
Desktop GPU	0.016	0.034	0.07
P30	0.16	5	1.9
Mate20	0.29	4.6	1.7
Pixel7	0.31	5.2	1.4

Table 3.3: *Runtime analysis (second/instance).*

Since VibVoice can effectively reduce the WER of AirPods Pro by 50%, we envision that VibVoice can be extended to other earphones.

User movements

We further evaluate the performance of VibVoice when the user is moving. We ask volunteers to move in the room with a speed of around 1 m/s. The result shows the improvement for a still volunteer is 58.3 %, while the same moving volunteer is 57.1%. The standard deviation for the still and moving case is 29.9 % and 30.4%, respectively. Although the improvement ratio slightly drops, VibVoice still preserves its performance under regular movements.

3.5.7 Runtime Evaluation

We test the execution latency of VibVoice and the two baselines (i.e., FSN (Hao et al., 2021) and SN (Tagliasacchi et al., 2020)) on a desktop PC (i.e., i7-11700k CPU and RTX 3060 GPU) and three smartphones (i.e., Huawei P30, Huawei Mate20, and Google Pixel 7). We run the inference of 5 seconds clip 100 times and record the mean latency. The results in Table 3.3 show that VibVoice reduces up to 31× and 12× less latency on average than FSN and SN, respectively. On the other hand, the results show that VibVoice exhibits a greater advantage in runtime for low-end devices. In conclusion, VibVoice can support real-time voice applications with a delay of fewer than 0.6 seconds (i.e., two times the inference latency) for processing 5 seconds audio clip. The real-time factor is only 0.12, significantly less than the minimal requirement, which means less energy consumption and space to process other tasks.

3.6 User Study

We recruit 35 volunteers to study the real perceived experience of VibVoice.

Questionnaire Design. We play a few recordings from the dataset described in Section 5.1 to volunteers. The length of each sentence is clipped or padded to 5 seconds. The first part is to evaluate the intelligibility of reconstructed speech. The participants are asked to listen to the audio clips enhanced by VibVoice and write down the sentences. We use WER to evaluate the results. In the second part of the study, participants listen to two audio clips with the same content and choose the audio with better quality in their perception. This study evaluates whether the reconstructed speech improves the user experience or not. We conduct two kinds of comparisons five times. One type is to ask participants to choose between audio enhanced by VibVoice and the original noisy audio. The second kind is to ask participants to choose between audio enhanced by VibVoice and the baseline (i.e., FSN). We use the correct ratio $\frac{P}{P+N}$ to represent the improvement of VibVoice, in which P and N are the numbers of choosing VibVoice and negative versa vice.

3.6.1 Study Result

According to the results of the first part, VibVoice achieves an overall WER of 21.5%, which is acceptable for understanding the audio content and confirm the effectiveness of VibVoice. According to the answers to the second question, the survey results show that 87% of the participants choose VibVoice over the baseline, and 72% of them choose VibVoice over the original audio without any enhancements. In addition, we discuss with participants why they prefer the original audio sounds over the baseline. The baseline can produce acoustic artifacts and sometimes wrongly suppress the sound of the target speaker. We note that some participants observed that the impact of artifacts and suppression is hindered after knowing the content or listening repeatedly. However, the speech generated by the baseline causes lots of misunderstanding for the first-time listener. In conclusion, the user study results show that VibVoice can enhance speech quality and improve user experience compared with the original audio and the baseline.

Chapter 4

EgoLog

4.1 Introduction

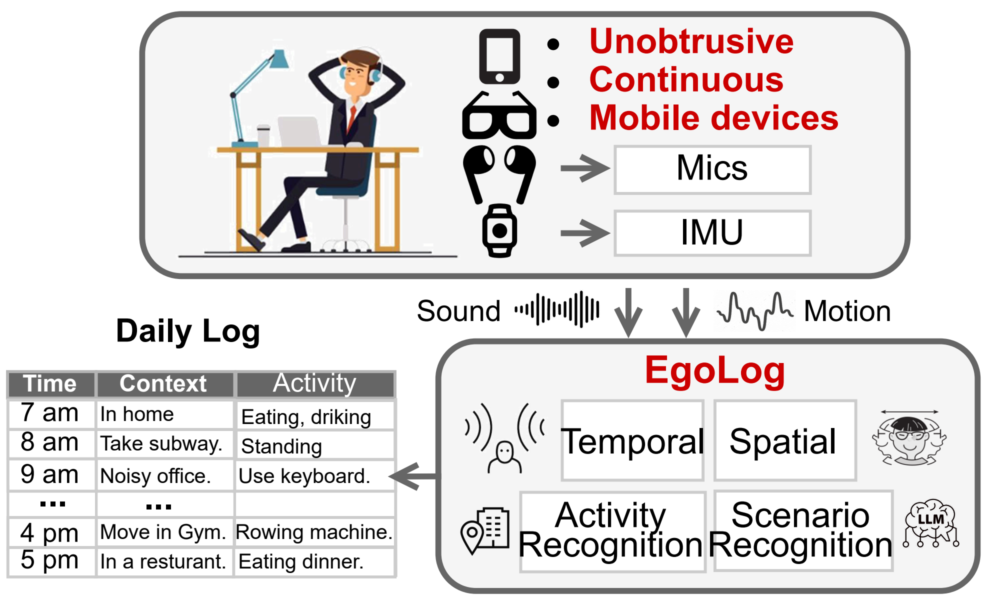


Figure 4.1: *EgoLog leverages microphones and an IMU on commercial mobile devices for both activity and scenario sensing.*

Human activity recognition is a critical tool for monitoring physical and mental health, which enables numerous applications such as personalized healthcare and improving health outcomes (Alam and Roy, 2014; Gedam and Paul, 2021). These days, smartphones and other mobile devices are practically extensions of ourselves, capturing real-time data about our daily routines. For example, motion data from an Inertial Measurement Unit (IMU) can be used for human activity recognition since it directly correlates to the body movement. Various applications including walking, standing, sitting, lying (Reyes-Ortiz et al., 2015) are developed with IMU. Besides, the built-in microphones can record audio data to support acoustic scene recognition (Martín-Morató et al., 2021), which is also relevant to human activity.

However, existing HAR methods on mobile devices are limited to coarse-grained, scenario-constrained detection. For instance, step counting is a common application, but it fails to capture diverse activities such as cycling, swimming, or strength training. Although prior work has extended sensing to vital signs, gait analysis, and exercise recognition, these approaches often focus on isolated tasks rather than continuous, holistic daily monitoring.

This paper aims to design a mobile sensing system for ubiquitous wearables, such as earphones and smart glasses, to enable the continuous tracking of detailed egocentric human activities. Our approach involves modeling daily life from the user's perspective (i.e., an egocentric view), moving beyond the recognition of consecutive but isolated activities typically collected in laboratory settings. We observe two key distinctions between fine-grained daily logging and conventional activity recognition: 1. Daily life comprises long-term, continuous activity that contains richer information than short-term activity, which is called "scenarios". For instance, a user may spend an entire afternoon at the office, engaging in numerous interrelated activities. 2. Daily life inherently involves interactions with the environment. Consequently, sensors like microphones capture a complex auditory scene comprising sounds from the user, the environment, or a combination of both. While these environmental inputs can provide valuable contextual information to benefit activity recognition, they may also introduce significant noise.

Several types of mobile devices are suitable for this task, including earphones, glasses, smartphones, and smartwatches, which users typically wear throughout the day and which possess rich sensing capabilities. The available sensors can be roughly divided into two categories: visual and non-visual. Visual sensors, such as cameras, can capture a first-person view rich with information. However, they are less practical for daily logging due to several limitations: 1) a limited field of view (FOV), 2) privacy concerns, and 3) high energy consumption for continuous sensing. Instead, we envision incorporating non-visual sensors such as motion and audio sensors for our purpose. Those sensors are widely available on different mobile devices, including but not limited to smartphones, glasses, smartwatches, and earphones. As Fig. 4.1 shows, EgoLog can continuously monitor a user's daily activities in an egocentric manner and generate a detailed daily log, both the short-term activity and long-term scenario, with several examples illustrated in the figure.

However, there are some unique challenges below:

- Existing mobile devices HAR models (Bhattacharya et al., 2022; Gong et al., 2023) lack the awareness of scenario information present in the real world. Specifically, user activities are driven by underlying reasons, and their distribution varies depending on the "scenario." For instance, in an "indoor cooking" scenario, the "jogging" activity is less likely to happen.

- HAR models that depend on audio recordings are susceptible to interference due to the lack of spatial understanding. As a result, the audio data may lead to potential errors, which can be harmful for post-processing. For example, when detecting a "clap" activity based on sound, a nearby person's action could also trigger the model.

- Multi-modal fusion of audio and IMU is challenging and may not outperform the uni-modal model (Huang et al., 2022). The heterogeneity of sensors indicates a weak correlation between them, where the effective fusion is particularly challenging and validated in MMG-Ego4D (Gong et al., 2023). Therefore, effectively extracting and leveraging sensor correlations is crucial for successful multi-modal fusion.

Towards resolving the above challenges, we propose the following designs of *EgoLog*:

First, we recognize the long-term scenario through temporal understanding, as detailed in Sec. 4.4.1. This is achieved using a two-stage model: In the first stage, contrastive learning is employed to project audio and IMU features into a shared space, identifying well-aligned segments as informative key frames. The second stage then aggregates these key frame features, along with the original unimodal features, over time using a sequence model to achieve robust scenario recognition.

Secondly, we distinguish user-generated activities from environmental sounds (including those produced by other people) in Sec. 4.4.2 by spatial understanding. Specifically, we formulate this as a sound localization task, where near-field sounds are treated as user-centric events, while far-field sounds are regarded as ambient background. To this end, we introduce a sound localization approach enhanced by natural human movement, which classifies sound sources into eight spatial regions. Our proposed motion transformer incorporates IMU readings to dynamically compensate for and refine localization estimates. This motion compensation module not only corrects errors induced by user motion but also alleviates common ambiguities in spatial perception, such as near-far confusion.

Third, we introduce an improved multi-modal network for activity recognition that integrates audio, IMU, and scenario information in Sec. 4.4.3. Unlike short-term sensor data (e.g., audio and IMU), the derived scenario captures long-term behavioral patterns and provides high-level semantic cues that significantly enhance recognition accuracy. The network employs dedicated unimodal backbones for feature extraction and leverages a self-attention mechanism to dynamically fuse multimodal inputs, where the scenario information is encoded into a dense representation via a text encoder.

Finally, in Section 4.4.4, we introduce a novel edge-cloud framework assisted by a Large Language Model (LLM) to further enhance scenario recognition. This approach leverages the strong reasoning capabilities of LLMs to support a smaller-scale local model. Specifically, we convert multi-modal data, including audio, IMU signals, and the outputs from both sound localization and the local scenario recognition module, into contextual information for the LLM. When the local model's prediction confidence is low, the edge device queries the cloud-based LLM for inference. The refined scenario output generated by the cloud is subsequently downloaded to fine-tune the local model, thereby improving its performance over time.

We implemented and evaluated EgoLog on three large-scale public dataset (e.g., Ego4D (Grauman et al., 2022)) and a self-collected dataset, which was collected by various form-factor from 10 participants, 12 activities for validation. We demonstrate that EgoLog obtains 12% improvement on activity recognition and 15% improvement on scenario recognition with low computation cost against strong multi-modal baseline (e.g., ImageBind (Girdhar et al., 2023)). The contributions of this paper can be summarized as follows:

- We propose a new audio-IMU fusion method from two perspectives: temporal and spatial understanding. We extract scenario-level features through contrastive learning and aggregate them from short-term windows for scenario recognition. Besides, by integrating motion information with sound source localization, we boost the performance with movement compensation.

- We propose a novel HAR method that takes both the scenario context and raw sensor data as input. On one hand, we distinguish similar activities by leveraging the extracted scenario information. On the other hand, we use the sound source and audio-IMU correlation to differentiate the user-generated sounds from background sounds.

- We propose an LLM-assisted scenario recognition paradigm. In this framework, processed sensor data is fed into an LLM for high-level reasoning. The LLM's output is then used to transfer actionable knowledge back to the local device for on-device inference.

- We evaluate EgoLog on both the public dataset and the self-collected dataset, and conduct a user study on whether the generated summary can be helpful in daily life. Evaluations of EgoLog on activity and scenario recognition demonstrate a performance improvement of up to 15% over the best of our baselines.

4.2 Related Work

4.2.1 Motion Sensing for HAR

IMU sensors in wearables and earables facilitate position tracking and user motion recognition, with their performance significantly impacted by sensor placement. For example, LIMU-BERT (Xu et al., 2021) recognizes up to six activities with IMU recording of a smartphone. Similarly, IMU on the wrist (e.g., smartwatch) can be used to recognize the basic activities (Chan et al., 2024). Current motion sensing (Chan et al., 2024) only supports coarse-grained tasks, even with a large-scale dataset (>3000 hours), such as physical activity level (4 levels) or activities of daily life (10 activities). Compared to smartphones and smartwatches, earables are a more reliable choice due to their relatively fixed position (Lyu et al., 2024; Han et al., 2023d,c, 2025), as we illustrated before. Nonetheless, they share the same limitations in fine-grained motion sensing.

To enhance motion sensing, deploying multiple IMU sensors can enable precise full-body keypoint estimation with up to 17 IMUs. The pose of humans can be used to recognize fine-grained human activity, as demonstrated by MotionBERT (Zhu et al., 2023). However, such an approach is impractical for everyday use. Instead, researchers propose leveraging commercial devices only, such as smartwatches, smartphones, and earables. For instance, combining multiple devices can enable fine-grained exercise activity recognition (Strömbäck et al., 2020). Similarly, IMUPoser achieves full-body pose estimation with just three IMUs (Mollyn et al., 2023). While using multiple devices improves performance, having all of them will have problems of high cost, user discomfort, setup complexity, and limited availability.

Another direction involves using pre-trained models to fuse IMU data with other modalities. Models like IMU2CLIP (Moon et al., 2023) and ImageBind (Girdhar et al., 2023) link motion data to text or vision, enabling zero-shot recognition via retrieval. However, their performance on fine-grained tasks remains limited. Recently, Large Language Models (LLMs) have shown promise in reasoning over motion data (Ji et al., 2024; Leng et al., 2024;

Xu et al., 2024b; Ouyang and Srivastava, 2024), often by converting sensor readings into text tokens for the LLM to process (Xu et al., 2024a).

In summary, prior work on motion sensing either employs multiple IMU sensors or remains confined to coarse-grained activity recognition. Since IMU sensing inherently lacks contextual or environmental information, which limits their ability to perform fine-grained HAR (Tong et al., 2020). Further advancements should focus on integrating additional modalities or external knowledge.

4.2.2 Acoustic Sensing for HAR

In addition to IMU sensors, acoustic sensors (microphones and speakers) are also commonly integrated into wearables and mobile devices. The use of a speaker allows acoustic sensing to be categorized into two approaches: active sensing (utilizing both the speaker and the microphone) and passive sensing (using only the microphone).

As the main research area of audio, there are multiple applications of passive audio sensing, including speech recognition and audio classification (Schmid et al., 2023). However, audio is related but does not directly reflect human activity. Instead, it relies on the sound caused by activity, which can be easily misled by a loudspeaker. As a result, audio-based HAR is often treated as a specialized form of audio classification (Cristina et al., 2024). In contrast, multi-channel passive sensing enhances spatial awareness by estimating the direction of arrival (DoA) of sounds using microphone arrays (Schmidt, 1986; Adavanne et al., 2018) or binaural earables (Yang and Zheng, 2022; Krause et al., 2021). While DoA provides useful context, it cannot confirm whether the source is human or differentiate the loudspeaker. To detect human presence, some approaches integrate feedforward microphones to capture vocal activity via head-related sounds (Zhang et al., 2024b), or use an IMU to capture the bone-conduction sound (He et al., 2023). However, they can only cover the vocal activity.

Different from passive sensing, active sensing analyzes transmitted acoustic waves and their reflections, enabling non-contact sensing. For the context of human activity, we are more interested in sensing not only the facial movements (Dong et al., 2024) with outward-facing speaker, such as upper body pose detection (Mahmud et al., 2023), and gesture recognition (Yang et al., 2024b). However, the speakers are usually oriented inward or designed to be effective only in the near field to maintain user privacy, which hugely limits the performance in coarse-grained recognition.

In summary, acoustic sensors in wearables enable rich information capture through either passive or active sensing. However, audio-based HAR is still limited, as passive sensing struggles to confirm human presence, and active sensing is constrained by practical considerations, necessitating multi-modal integration for improved performance.

4.2.3 Multi-modal Sensing for HAR

Multi-modal learning of audio and IMU data addresses limitations in motion sensing or acoustic sensing by leveraging the complementary strengths of both modalities. For instance, VibVoice (He et al., 2023) uses bone-conduction vibration and audio in earables to distinguish a user's voice from others. Similarly, (Min et al., 2018) employs IMU for motion detection and a microphone for conversation detection, processed independently. Other wearables, such as wrist-mounted or head-mounted devices, also utilize multi-modal approaches; SAMoSA (Mollyn et al., 2022; Bhattacharya et al., 2022) integrates audio and IMU on smartwatches to recognize up to 26 activities. However, multi-modal learning is not universally superior, as its effectiveness depends on factors like data quality, modality correlation, and model design. For example, the fusion of audio and IMU on a diverse dataset like Ego4D (Grauman et al., 2022) can be challenging. As demonstrated by MMG-Ego4D (Gong et al., 2023), where a multi-modal model for smart glasses underperformed compared to single-modal models in recognizing nearly 200 actions, highlighting that multi-modal approaches do not always guarantee improved performance (Huang et al., 2022).

Beyond integrating audio and IMU sensors, incorporating additional knowledge sources can enhance motion sensing. For example, SAMoSA (Mollyn et al., 2022) uses automatic

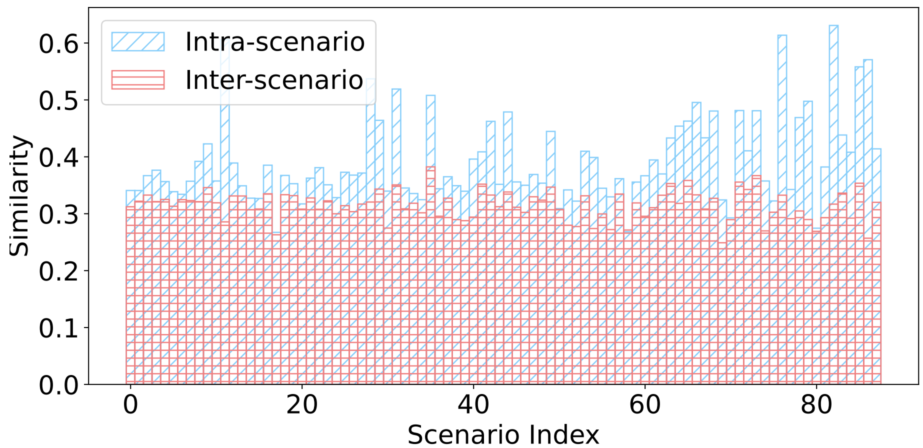


Figure 4.2: *Motivation for scenario understanding: The similarity of embeddings for intra- and inter-scenario activities.*

context detection to trigger context-specific classifiers, though limited to predefined classes. Furthermore, the advanced LLMs facilitate multi-modal processing, either at the prompt level, incorporating IMU, GPS, Wi-Fi, or Bluetooth data (Yang et al., 2025; Xu et al., 2024b; Yang et al., 2024a), or at the token level, combining audio and IMU (Han et al., 2023a; Girdhar et al., 2023). While promising, those works lack explicit multi-modal fusion, especially for the heterogeneous multi-modal data from mobile devices.

In summary, the fusion of audio and IMU can be challenging since the correlation between them can be weak. While incorporating external knowledge or using powerful models like LLMs for fusion can enhance recognition, current methods still lack effective techniques for deeply integrating the heterogeneous data.

4.3 Motivation Study

As discussed in Sec. 4.1, we mainly have three challenges towards a multi-modal HAR on mobile devices: 1) lack of scenario awareness, 2) being sensitive to ambient noise, and 3) fusion collapse. To explore these challenges and evaluate the potential for overcoming them, we conduct a series of experiments to analyze and demonstrate promising directions.

4.3.1 Scenario

Scenario information refers to long-time activities (e.g., playing with a pet or working at home) that reflect the user's real-world behavior. Capturing such scenarios in datasets is challenging, as it requires volunteers to perform activities naturally, mimicking real-life conditions. Generally, scenario is the ensemble of multiple related and continuous activities with a realistic distribution, reflecting meaningful interactions with the environment.

To investigate the impact of scenario information on HAR, we utilize the Ego4D dataset (Grauman et al., 2022), which captures real-world activities through smart glasses equipped with IMU and audio sensors. The dataset provides annotations at both the window level (every 10 seconds) and the video level, represented as plain text. We treat window-level annotations as activity labels and video-level annotations as scenario labels, reflecting the broader scenario of prolonged activities. To examine the properties of scenarios, we employ Sentence-BERT (Reimers, 2019) to generate text embeddings for activities across 91 scenarios. We then compute activity similarity both within scenarios (intra-scenario) and between scenarios (inter-scenario). Intra-scenario similarity measures the similarity of activities within the same scenario, while inter-scenario similarity compares an activity from one scenario to a randomly selected activity from a different scenario.

The results are shown in Fig. 4.2, revealing that the similarity within a scenario (intra-scenario similarity) is over 20% higher than the similarity between different scenarios (inter-scenario similarity). This suggests that each scenario has a distinct distribution of activities. Specifically, a scenario effectively represents the user's daily log since it encompasses a longer time frame. Additionally, we notice that the inter-scenario similarity is generally above 0.3, indicating that many basic activities (such as sitting and walking) are common across all scenarios. Therefore, it is crucial to eliminate the influence of these basic activities in scenario recognition, as they are present in every scenario.

In summary, the study on the in-the-wild dataset indicates unique activity distributions per scenario, which can be leveraged to recognize scenarios by identifying scenario-specific activities.

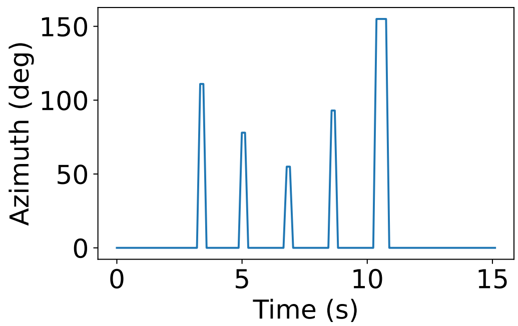


Figure 4.3: *Sound localization (Schmidt, 1986) under motion*

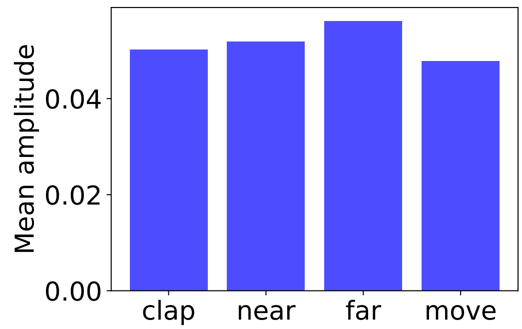


Figure 4.4: *Sound volume may not related to distance.*

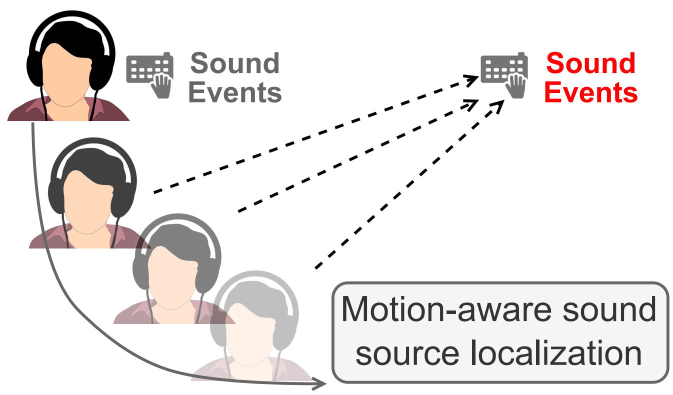


Figure 4.5: *Motivation for spatial understanding: estimate the distance of a sound event by motion.*

4.3.2 Sound Location

In practice, recorded audio often includes far-field sound sources that are unrelated to the user's activities, underscoring the importance of sound localization for effective HAR. Wearables that are consistently worn on the user's head provide a stable and reliable position for capturing detailed ambient sound information. Additionally, modern wearables feature multi-channel recording capabilities, enhancing their spatial sensing potential. In the following, we divide sound localization into direction and distance estimation, which we discuss as follows.

The estimation of sound direction is one of the mature, basic, but still challenging applications of a microphone array. The accuracy highly depends on the number of microphones, which is unfortunately not sufficient for most commercial devices. Considering the most common setting of stereo recording, there are mainly two problems: the deviation of direction (especially for front-back confusion) and estimation of sound distance, as illustrated in Fig. 4.5. We envision that the above challenges may be solved by motion compensation. Specifically, we can aggregate the consecutive sound localization results along with the movement of the user (microphone) to refine the results or even get the distance.

To illustrate them, we positioned a sound source in front of the user and utilized a binaural microphone along with the algorithm from (Schmidt, 1986) to estimate the direction of arrival, as shown in Fig. 4.3. The user is instructed to move the body slightly and naturally. We found that the estimation was not satisfactory, as it fluctuated from the true value of 90 degrees. To validate whether sound volume is enough for differentiating near and far, we conducted an experiment where the sound source was at different distances from the microphone. As shown in Fig. 4.4, there was no notable difference in volume between different cases, indicating that volume alone is an unreliable indicator for distance. The final bar in Fig. 4.4 reveals that user movement slightly affects amplitude. Moreover, the presence of multiple sound sources can result in an even more complex mix of random sound volumes.

4.3.3 Multi-modal Fusion

To investigate scalability bottlenecks in activity recognition using wearables, we trained unimodal supervised models with audio and IMU data on the Ego4D episodic memory subset (200 activity classes). We used EfficientAT (Schmid et al., 2023) for audio and LIMU-BERT (Xu et al., 2021) for IMU as feature extraction backbones, comparing them to a multi-modal model via late fusion. Activities were clustered into varying class numbers using Sentence BERT (Reimers, 2019) embeddings and K-Means (MacQueen, 1967), following MMG-Ego4D (Gong et al., 2023). Results (Fig. 4.6) show a significant performance drop as class numbers increase, indicating challenges in recognizing diverse activities with audio

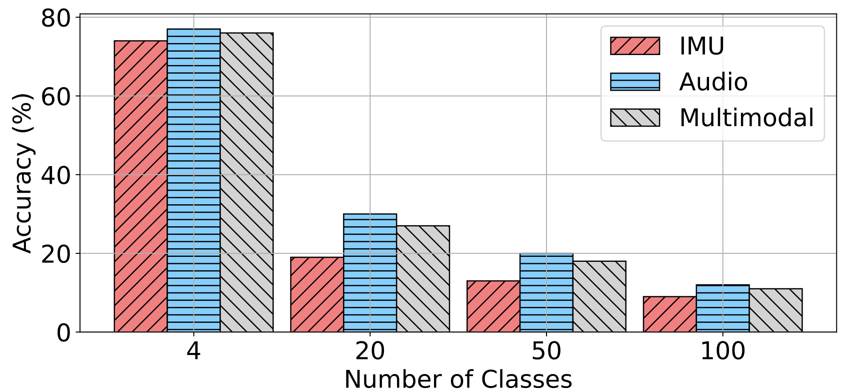


Figure 4.6: *The performance of an existing activity recognition with different class numbers and modalities.*

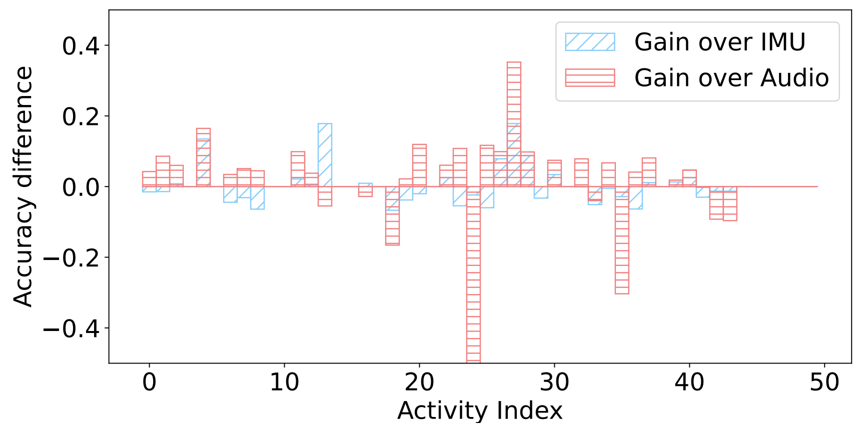


Figure 4.7: *Performance gain from multi-modal fusion.*

or IMU alone. Multimodal fusion often underperforms unimodal models, suggesting ineffective extraction of complementary features at larger scales.

Previous work (Mollyn et al., 2022) demonstrates that multi-modal learning can outperform the unimodal model due to the complementary nature of each other. However, it may not apply to all cases, where another study (Gong et al., 2023) finds multi-modal learning with audio and IMU on Ego4D leads to even worse performance than the single-modal solution. To investigate the reasons behind the unexpected results, we compared the accuracy of multi-modal and uni-modal models for each class, as shown in Fig. 4.7. We observe that the advantages of multi-modal fusion fluctuate for both modalities. This indicates that supervised multi-modal learning can not automatically select the right modality, failing to benefit from modality fusion.

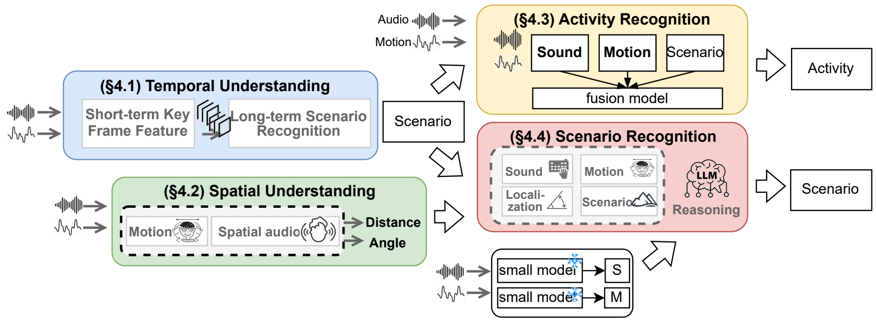


Figure 4.8: *The system overview of EgoLog: the temporal and spatial understanding are two modules for multi-modal fusion, providing important input to the later modules to recognize activity and scenario*

To investigate the effectiveness of LLMs in alleviating the problem, we employ the ImageBind-LLM (Han et al., 2023b), which converts both modalities into features and puts them in front of text embeddings. We instruct the model by the following prompt: "<sensor> According to the audio and IMU reading on the earphone of the user, what is the activity of the user?", while <sensor> refers to the place of sensor features. Since the LLM can have random output, we also add a prompt to force it to output one of the potential activities. To reduce complexity, we selected the top 30 activities in the subset of Ego4D (episodic memory). The BERT score of ImageBind-LLM is 0.04, 0.07, and 0.05 for precision, recall, and F1-score, respectively. As long as the performance is too low, we inspect the output and find that the LLM may not fully understand the task (multi-modal fusion). Furthermore, we inspect whether fine-tuning the model can resolve the problem by training an adapter (Zhang et al., 2024a) for ImageBind-LLM. The adapted LLM achieves precision, recall, and F1-score for BERT Score of 0.47, 0.46, and 0.47, respectively. Although it is improved a lot, it is still much lower than usability and is far from practical.

4.4 System Design

Figure 4.8 provides an overview of EgoLog, a mobile system designed to capture egocentric daily logs over both long-term and short-term periods. Specifically, Section 4.4.1 and Section 4.4.2 detail the fusion of audio and IMU data to achieve a deeper understanding of scenario (i.e., long-term descriptions) and spatial information (i.e., the relative location of

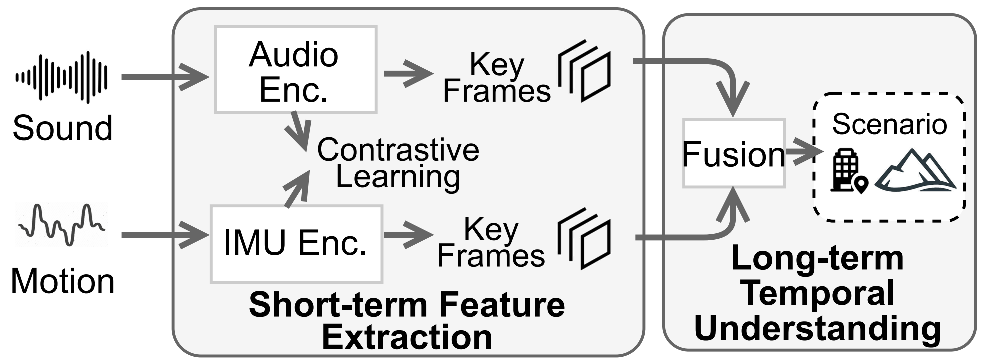


Figure 4.9: *Temporal understanding: we use contrastive learning to extract the cross-modal feature, which is important for the temporal-level fusion that recognize the scenario.*

sound sources). These insights serve as foundational clues for subsequent processing. In Section 4.4.3, we integrate the scenario information into the multi-modal activity recognition process, effectively mitigating modality collapse. Finally, Section 4.4.4 describes the use of an LLM as an expert to refine the scenario recognition from the previous stage; the resulting knowledge is subsequently transferred back to the local device.

4.4.1 Temporal Understanding

The results of our measurement study indicate that recognizing a large number of activities using IMU or audio data with an end-to-end model is challenging. Therefore, we propose a scenario recognition approach to provide a prior information that can assist in activity recognition (as detailed in Section 4.4.3). Additionally, scenario recognition itself is valuable for gaining a better understanding of daily logs. As previously discussed, a scenario can be considered as a long-duration activity composed of multiple correlated sub-activities. To achieve accurate recognition, it is essential to develop an improved method for temporal analysis. The pipeline is illustrated in Fig. 4.9.

Problem analysis

The audio and IMU data represent their own characteristics and different advantages. As a result, the pretrained encoders tailored for these modalities are unique and fine-tuned for particular objectives. This variation suggests that simply merging the output from these encoders might not produce the optimal outcomes, as corroborated in Sec. 4.3. Thus, it is crucial to thoughtfully integrate these diverse modalities to fully leverage their collective advantages.

We aim to tackle scenario recognition, which refers to identifying high-level human activities over an extended period (over 30 seconds). As detailed in Sec. 4.3, current pre-trained encoders already hold some scenario-level information, although they still fall short of ideal performance. A straightforward approach involves training a classification model using a dataset annotated with scenarios in an end-to-end fashion. Nonetheless, we have identified two persisting challenges: (1) the end-to-end model may struggle to generalize across different input lengths. (2) The performance might be inferior if scenarios are not included in the training data. Therefore, we propose training a new encoder for two modalities to obtain scenario-level features from brief time windows, which will then be applied in a secondary scenario recognition phase.

Given audio data and IMU data, the corresponding annotations for activity and scenario are assigned. Ideally, we could estimate activities individually and recognize scenarios accordingly. However, many activities may not relate to the audio or IMU data, such as silent motions that exceed the audio's capability. Instead, we propose using two encoders: one for audio and one for IMU data. Their outputs would project the same feature space, representing features of the same scenario. Like activities, the two modalities may not always directly correlate with the scenario, so the mapping can fail sometimes. To address this, we can estimate the scenario by combining multiple windows of data, compensating for the sparsity of features.

Contrastive learning

Based on the preceding analysis, employing contrastive learning is optimal for capturing scenario-level features. Contrastive learning focuses on acquiring low-dimensional representations (SE_i) by differentiating between similar and dissimilar data samples. The method aims to group similar samples closely in the representation space while distancing dissimilar ones using the Euclidean distance. The efficacy of contrastive learning has been demonstrated in earlier research (Han et al., 2023a; Girdhar et al., 2023). Within our framework, similarity and dissimilarity can be defined in two ways. Firstly, any data point aside from itself is deemed dissimilar. Secondly, data points within the same scenario are viewed as similar, whereas others are regarded as dissimilar. We expect that the first approach will yield more robust features, since it does not rely on explicit scenario annotations, and this will be assessed during the evaluation.

Regarding the network design, we employ two distinct deep neural networks to derive embeddings: LIMU-BERT (Xu et al., 2021) for the IMU data and EfficientAT (Lyu et al., 2024) for the audio input. Each sample spans a two-second window. The training of both encoders is based on the cross-entropy loss function, structured as follows:

$$logits = (Audio_e \cdot IMU_e^T) \cdot \exp(\tau),$$

where $Audio_e$ and IMU_e refer to the features from two encoders, and τ refers to the learned temperature parameter (Chen et al., 2020). Then, we get the loss using cross-entropy loss as follows:

$$L = cross\_entropy\_loss(logits, label, axis = 0).$$

Scenario recognition

Assuming we extract the features from $Audio_e$ and IMU_e by contrastive learning, they may not immediately convey the scenario as previously noted. We observe that a robust example in which both modalities relate to the scenario will exhibit a strong similarity between $Audio_e$ and IMU_e. This high similarity indicates that both modalities are effectively mapped

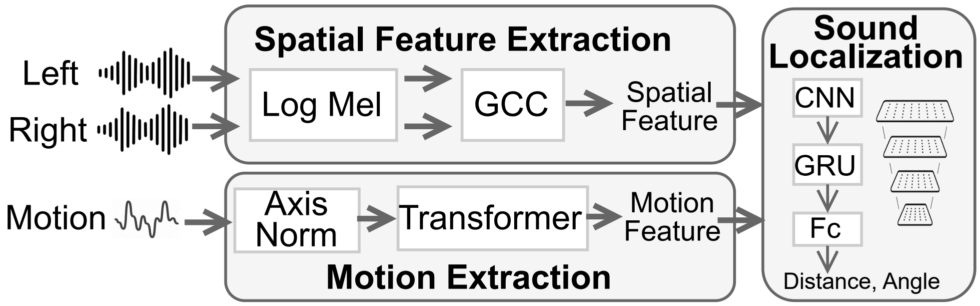


Figure 4.10: *Spatial understanding: we differentiate sound by location by the combination of spatial audio and motion, leveraging the natural but random movement of the user.*

into the same feature space, highlighting their importance as key features. Conversely, low similarity suggests that the two modalities have not been successfully integrated into a unified representation of the scenario. Ideally, we would only keep frames with high similarity. However, the dynamics of execution can be problematic, introducing instability. Furthermore, it is difficult to set a definitive threshold to classify all frames below it as "useless." Given the inherently weak correlation between audio and IMU, examples with high similarity may be rare. Therefore, it is important to leverage as much information as possible from the available data. To address this problem, we propose extending the key features along the temporal dimension. We integrate a sequence model—either an LSTM or a transformer—to compile features over time (as shown in the right section of Fig. 4.9). This two-stage architecture enables the model to capture both short-term and long-term dependencies within the features, thereby improving the accurate recognition of extended scenarios. Finally, a linear layer followed by a sigmoid activation function is used to predict the probability for each scenario.

4.4.2 Spatial Understanding

Existing activity recognition methods by audio are often misled by environmental events. As illustrated in Sec. 4.3, it is not enough to use sound volume to differentiate the near-field and far-field sound. Here, we are interested in the near-field sound because of the close correlation between it and the user's interaction with the environment.

Sound localization

The location of sound can be estimated by a multi-channel microphone array. Specifically, the time of arrival of the sound source for one microphone is: $t_i = \frac{\sqrt{(x_s - x_i)^2 + (y_s - y_i)^2}}{c}$, where the location of microphones $\mathbf{p}_i = (x_i, y_i)$, $i = 1, \ldots, N$, and a sound source at $\mathbf{s} = (x_s, y_s)$. Since we are not aware of the start time of the sound source, we can only obtain the Time Difference of Arrival (TDoA) between two microphones like i and j instead. Considering there are only two microphones for binaural earables, we can only have one equation that is less accurate and can't solve front-back confusion.

Building on the work of SELDNet (Adavanne et al., 2018) and DeepEar (Yang and Zheng, 2022), we propose a neural network-based binaural localization. Overall, our model performs an 8-class classification, distinguishing between four directions (front, left, right, back) and two distance categories (near and far). The binaural audio input, is converted to two log-Mel spectrograms and generalized cross-correlation coefficients of spectrograms, combined along the channel dimension to form spatial features with shape $B \times 3 \times T \times 64$, where B is the batch size and T is the number of frames. We process the audio feature with three convolutional layers, transforming it to (B, 128, T/5, 64). Then it is followed by a GRU layer, multi-head attention (resulting in B, 128, T/5, 64), and a linear layer that outputs (B, T/5, C).

Movement compensation

As illustrated in the motivation study, user movement can be leveraged to enhance spatial understanding. While similar concepts have been implemented in prior works such as (Zhu et al., 2021; Seo et al., 2022), these approaches require either explicit user gestures or a fully controllable robot. In contrast, our proposed EgoLog operates without relying on predefined gestures or manual intervention, as it utilizes natural, unintentional body movements that occur in daily life.

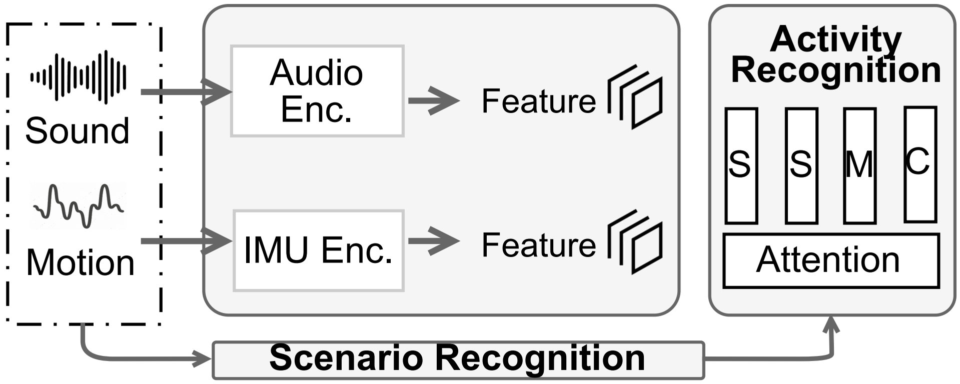


Figure 4.11: *Multi-modal activity recognition: We extract features from each individual modality (uni-modal features), combine them with the scenario information derived from temporal understanding, and merge them at the token level for final activity recognition.*

To realize this, we introduce an end-to-end method where IMU data—structured as (B, 6, T) and comprising both accelerometer and gyroscope measurements—is processed through two transformer layers. This module outputs a representation of shape (B, 384, T), which is then passed through a linear layer to produce an output of size (B, T/5, C). The resulting features are subsequently concatenated with the output from the GRU module. The overall model performs an 8-class classification (C=8), distinguishing between four spatial directions (front, left, right, back) and two distance categories (near and far).

In the scope of EgoLog, we can not assume that only one sound source exists. Instead, we may encounter the case where there are multiple sound sources overlapping. Since both sound localization and audio recognition can be multi-label, the problem becomes an association confusion, and we will discuss how to leverage the information in a later section.

4.4.3 Scenario-aware Activity Recognition

The methods outlined in Sec. 4.4.1 and 4.4.2 offer a promising path for audio-IMU fusion. Next, we propose to integrate them into HAR.

Our proposed multi-modal network consists of two main components: unimodal backbones and a fusion module. The unimodal backbones include separate feature extractors that gather features from each input modality. The fusion module is designed to combine and aggregate these features from the unimodal backbones, resulting in a fused output that denotes the activity of the user. Although the design described above shows promise, it shares similar input and output characteristics with those presented in Section 4.3, which has already been validated. As a result, it may not perform well in human activity recognition due to a lack of multi-modal correlation.

There are two common approaches for fusing modalities: either use a multi-layer perceptron to process the concatenated representations of different modalities, or utilize a Transformer-based fusion module that processes a series of tokens from various modalities (Gabeur et al., 2020). We have chosen the transformer-based fusion design because it can easily scale to accommodate any number of input tokens through attention mechanisms. This flexibility is particularly important, as our multi-modal model needs to handle data with a varying number of modalities (more than only audio and IMU) for the task. Specifically, suppose the scenario information is a feature with the same dimension as other modalities, the transformer module can easily integrate it. The final output of the fusion module is calculated by averaging the output tokens, rather than relying on the CLS token. The formulation of the token for one modality Z_m is below:

$$Z_m = (Encoder_m(S_m), e_m), \quad Z = f(Z_1, Z2, Z_3...)$$

, where m represents each modality, which includes audio, IMU, and text. The term e_m denotes the learnable embedding specific to each modality. The transformer fusion module will combine multiple Z_m tokens and produce a unified Z feature with consistent dimensions. Similar to the previous section, we keep the audio and IMU encoder used for scenario recognition since both of them are lightweight and can be deployed on a mobile device. By default, we use the standard transformer with a number of layers of 2.

Next, we extend the multi-modal model that can handle not only the audio and IMU, but also other auxiliary information, such as scenario:

- Specifically, we collect scenario information from Sec. 4.4.1 in textual form. To integrate it, we use the ImageBind (Girdhar et al., 2023) text encoder to map this information into the same feature space as the other modalities. Note that the text embeddings can be precomputed and stored to reduce computational overhead for each scenario. When multiple scenarios are detected, we concatenate their corresponding text representations.
- For spatial information from Sec. 4.4.2, we treat it as an indicator for interference detection. When far-field sound is detected with a confidence score exceeding 0.9, the audio recording is considered unreliable. In such cases, we can either discard the audio recording and rely solely on motion sensing or apply distance-based sound extraction (Chen et al., 2024) to retain only the near-field sound.

4.4.4 LLM-assisted Scenario Recognition

As discussed in Section 4.3, LLMs may be suboptimal for direct activity recognition. However, we posit they are well-suited for higher-level scenario recognition, as longer data sequences provide a richer information base. This aligns with the findings in Section 4.4.1, where longer sequences also benefit multi-modal fusion—a property we expect to transfer effectively to LLM reasoning. As illustrated in Fig. 4.12, our approach involves formatting sensor data (five seconds window) into text and using tailored prompts to guide an LLM in scenario recognition. The LLM's output can further refine local scenario predictions. It is important to note that all LLM processing is performed on the cloud, ensuring that EgoLog remains a lightweight mobile system.

Prompt design

Existing studies primarily leverage the reasoning capabilities of LLMs on text-like data. This subsection introduces the fusion of multi-modal signals from audio and motion sensors for LLM-based reasoning in EgoLog, as illustrated in Fig. 4.12. Instead of re-training the LLM or adding an adapter, we focus on designing a specialized prompt to inject these

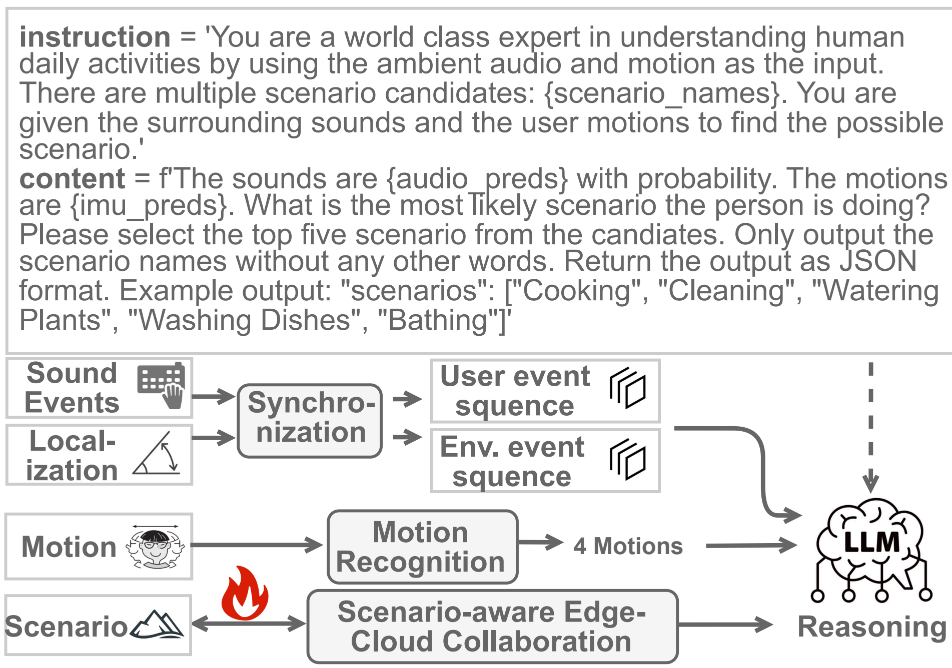


Figure 4.12: *LLM-assisted scenario recognition: Utilizing prior outputs, we differentiate sound events into those originating from the user and those from the environment. We then combine it with motion and scenario recognition results and feed it into an LLM for reasoning (the upstream process). The LLM's output is then used as a pseudo-label to re-train the local model (the downstream process).*

multi-modal contexts. Specifically, we construct a demonstration within the prompt to guide the LLM in reasoning with such sensor data. The prompt is structured as follows: “You are an expert in human activity analysis. You will receive audio and IMU recognition results, including extra information on potential scenarios. Using this information, reason and refine the person’s scenario. The scenario must be one of the following categories: [List of categories]. Your output must be a single line containing just one word or phrase.” This prompt, along with the formatted input data as shown below, is fed to the LLM.

Audio. We employ a pre-trained audio recognizer (Schmid et al., 2023) to transcribe audio into text by selecting the top three predictions. The output is structured as "the detected audio classes are <class1, prob1>, <class2, prob2>, <class3, prob3>...", where each entry is referred to as a sound event. To minimize noise, we retain only the top five predictions with a confidence score exceeding 0.3. Since the classifier’s label set is based on the AudioSet ontology, the defined classes are highly fine-grained, often resulting in multiple similar classes with comparable probabilities. This lack of discriminative power hinders effective reasoning. To address this, we apply a two-level hierarchical clustering of the classes to their parent categories, reducing the total number of classes from 416 to 50 and improving generalization.

IMU. Following the IMUCLIP approach (Moon et al., 2023), we classify IMU data into four activities: walking, moving, standing up, and sitting down. While previous works have demonstrated the ability to recognize more fine-grained activities using IMU data, their generalization capabilities remain questionable due to reliance on scenario-constrained datasets.

Spatial. We leverage spatial information from Sec. 4.4.2 by applying distinct processing strategies to near-field and far-field audio inputs, guided by tailored prompts. For near-field sounds, we use the prompt: “the <sound event> is happening in the near front of the user, which is likely to be related to human activity.” For far-field sounds, the prompt is: “the

<sound event> is happening in the <sound location>." When both near-field and far-field sounds are detected, it is unclear which event is near or far. As a result, we can turn on the distance-based sound separation (Chen et al., 2024) that seperate the audio into two and perform the sound recognition, respectively. We leave the optimization of the separation in the future work since it is unrelated to the main body of EgoLog.

Scenario-aware LLM Assistance

We define an upstream and downstream collaboration paradigm where the edge device seeks assistance from a cloud-based LLM, rather than naively querying the response.

For the upstream path, we leverage the superior scenario recognition capabilities of LLMs, especially for out-of-domain data. The scenario estimated by the edge device can be treated as a noisy reference, providing valuable preliminary information for the LLM. We integrate this estimation, along with its confidence score, into the prompt supplied to the LLM. Specifically, the estimated scenarios are added to the prompt, augmented by the text: "the preliminary scenario estimation is <scenario, confidence>." However, frequent queries to the cloud service raise concerns about cost, privacy, and latency. Ideally, all modules of EgoLog should be executable on a mobile device if the performance is sufficient. Consequently, we propose a collaborative framework that strategically uses the cloud LLM to enhance the local model. Specifically, a confidence threshold (0.5) is set for the local scenario recognition model. A cloud query is triggered only when the local model's confidence is low.

Complementing the upstream process, we also propose a downstream collaboration where the cloud LLM's output can conversely assist the local model. Specifically, the data points associated with uncertain cases (which were uploaded to the LLM), along with the LLM's refined results, are cached and used to fine-tune the local model, which enables continuous on-device improvement. Within the scope of EgoLog, we do not consider adding new class to the local model since the required data can be huge, which leads to a long time for waiting.

4.5 Evaluation

4.5.1 Experiment Setup

Implementation

We implemented our system on a Raspberry Pi 4, which was connected to multiple devices: binaural earables (MU6 (Mu6, 2020) and Sound Professionals (sou, 2015)), headphones, and smartglasses. These devices support multi-channel audio recording. Due to the lack of a commercial open platform with the same functionality, we prototyped our system using a 3D-printed model and a configurable microphone array, where the headphone prototype support 4-channel recording and the smart glasses support 4-channel, all devices are shown in Fig. 4.13. An IMU with a sampling rate of 200 Hz was attached to both the earables and the smartglasses. The neural network was trained on a server equipped with an Intel i7-11700k CPU and two Nvidia RTX 3090 GPUs using PyTorch. For inference, all computational tasks—except for summary generation—are performed on the smartphone.

Public dataset

To comprehensively evaluate EgoLog performance, we construct our dataset with the blow public datasets:

Ego4D: Ego4D (Grauman et al., 2022) is a large-scale dataset collected by various brands of head-mounted cameras (e.g., GoPro), offering synchronized multi-modal data (i.e., video, audio, and IMU signal). We consider the moment subset of the dataset, which consists of 158 classes. We can perform k-means clustering on the text embeddings to adapt to different levels of granularity. For the scenario annotation, we use the scenario annotation (Grauman et al., 2022) for each video since they have a similar definition. In total, there are 91 scenarios in the dataset. Considering Ego4D is the most diverse and large-scale dataset, we set it as the default dataset.

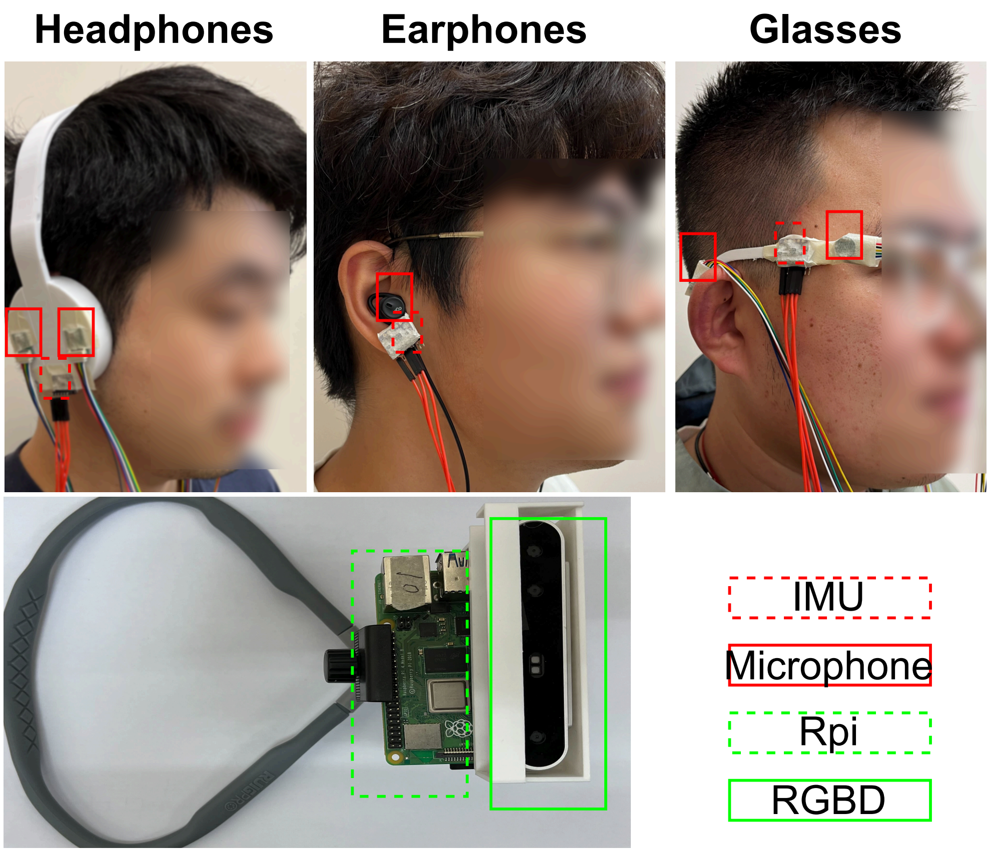


Figure 4.13: *Devices used in our data collection.*

EgoADL: EgoADL (Sun et al., 2024) is also a large-scale dataset collected in the real world with Android smartphones, including Wi-Fi, audio, and IMU data. In this paper, we exclude Wi-Fi data for fair comparison. We perform a similar clustering for the 221 classes of actions in the dataset.

SAMOSA: SAMOSA (Mollyn et al., 2022) is a relatively small-scale dataset collected by an Android smartwatch (i.e., Fossil Gen 5), including audio and IMU. There are 26 activities in total.

Self-collected dataset

The existing public datasets already include data from various devices—such as smart glasses, smartphones, and smartwatches—offering considerable duration and diversity. The purpose of our self-collected dataset is not to surpass these strengths but to address their limitations. Specifically, the spatial understanding discussed in Section 4.4.2 relies on spatial audio, which is absent from the aforementioned datasets but is accessible using commercial devices. To compensate for this gap, we collected a new dataset featuring spatial audio recordings from different form factors: glasses, headphones, and earphones.

We constructed a dataset in which a volunteer acted as the user and a loudspeaker served as an audio interference source. The loudspeaker played audio samples from the NIGENS dataset (Trowitzsch et al., 2019), which contains 14 types of common sound events. The user, in contrast, performed the following 12 activities: standing up, sitting down, clapping, walking, talking, drinking, typing on a keyboard, eating, watching TV, washing hands, cleaning a room, and writing. During data collection, the loudspeaker's position remained fixed. To simplify the experiment, the user's location was controlled for most activities (excluding walking), allowing us to manually record the relative position between the user and the loudspeaker. However, as users were not required to remain perfectly still, their precise position was not controlled with high accuracy. To ensure the dataset's validity, users were instructed to limit their positional variance to within the spatial resolution of our classification model. In essence, while the user's location was dynamic, the annotation was treated as static from the perspective of the classification task. For the walking activity, where the relative position changes continuously, we used an RGB-D camera to perform Simultaneous Localization and Mapping (SLAM) to estimate the user's trajectory. This estimation was based on the assumption that the user started from a fixed, known position in the room (the location of the loudspeaker is also fixed and known).

Baselines

We adopt the following evaluation metrics in this paper. For the multi-label classification (i.e., scenario and spatial), we use the multi-label F1-Score: $2 * TP/(2 * TP + FP + FN)$. For the single-label classification (i.e., activity), we use the accuracy as the metric.

To evaluate our approach, EgoLog, we have selected different baselines for activity recognition and scenario recognition, respectively. Specifically, we consider SAMoSA (Mollyn et al., 2022) and Imagebind (Girdhar et al., 2023) as our baselines for activity recognition, which are conventional supervised multi-modal model that accepts both audio and IMU. For the scenario recognition, both of the above baselines with larger data duration are considered. To ensure a fair comparison, all of them are fine-tuned with the same data.

4.5.2 Activity Recognition

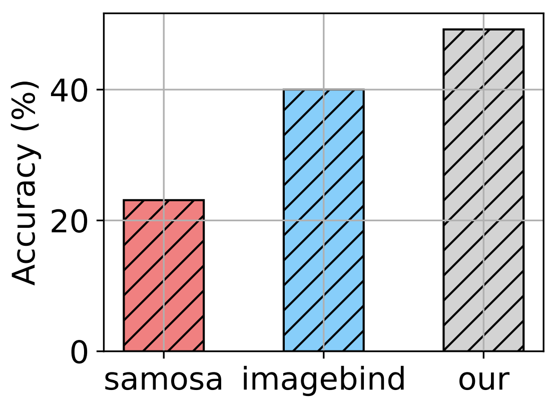


Figure 4.14: *Activity recognition on Ego4D.*

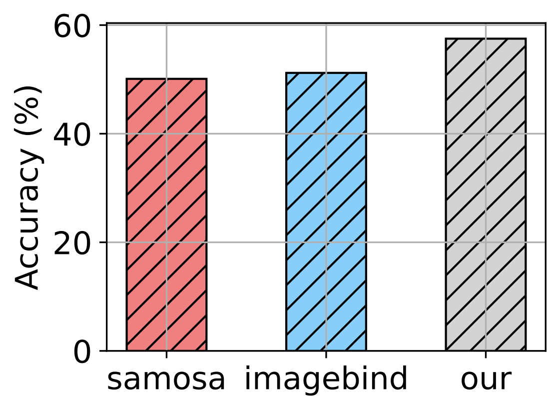


Figure 4.15: *Activity recognition on EgoADL.*

We present the performance of our method for human activity recognition on the Ego4D, EgoADL, and SAMOSA datasets. As shown in Fig. 4.14, 4.15, and 4.16, our approach consistently outperforms the baselines, with the most notable improvement on the SAMOSA dataset.

The performance gap is most significant on the Ego4D dataset (23% vs. 52%), which is the most challenging. Compared to ImageBind, EgoLog achieves superior performance

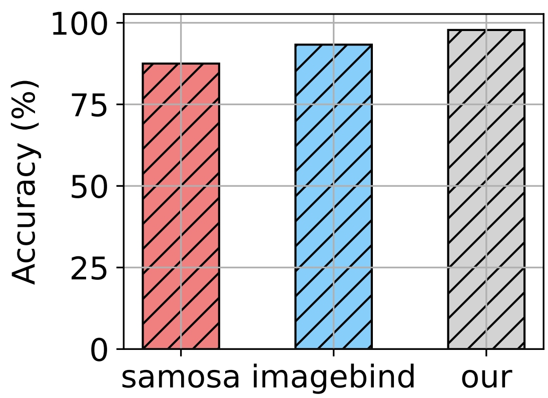


Figure 4.16: *Activity recognition on SAMOSA.*

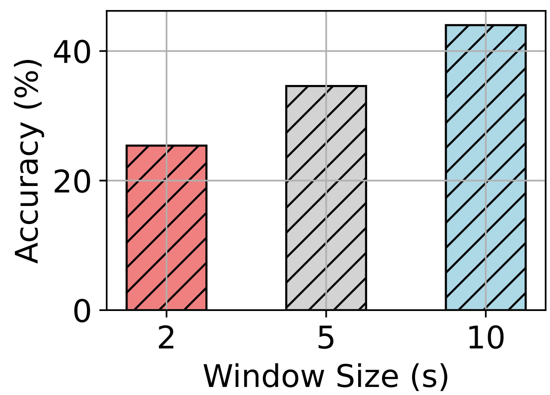


Figure 4.17: *Activity recognition vs. data length.*

(52% vs. 40%) with significantly lower computational cost.

On the other two, less diverse datasets, the advantages of EgoLog are slightly reduced. We attribute this primarily to a lack of contextual information; Ego4D data was collected "in the wild," while the others were gathered in moderately controlled experimental settings. Despite this, our approach still outperforms the best baseline by 6%on EgoADL and 4%on SAMOSA, demonstrating its robustness. Note that the absolute accuracy for SAMOSA is the highest, which is expected as it has the smallest number of classes, making it the least challenging dataset.

Data length. Since the length of the data significantly influences both audio and IMU data, we first establish the hyperparameters before evaluating the system. We present the performance of human activity recognition along with various data lengths, in Fig. 4.17. We observe that the HAR performance stays stable when the data length is larger than five seconds, while significantly dropping when the data length is one second. As a result, we keep the data length of HAR to five seconds.

Number of classes. We split the sampled Ego4D dataset into activities and evaluated their performance in Fig. 4.18. We observe that there are performance gaps between activities, indicating that the complexity of each of them is different. Specifically, we inspect the top

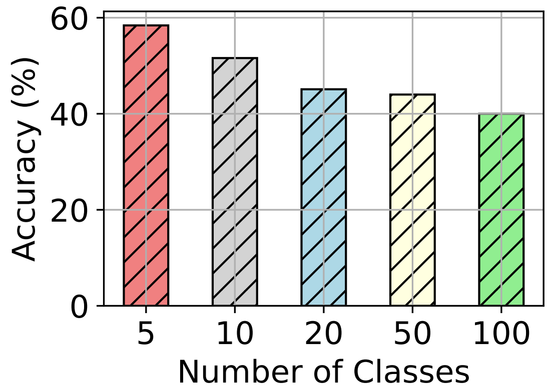


Figure 4.18: *Human activities recognition performance vs. number of classes.*

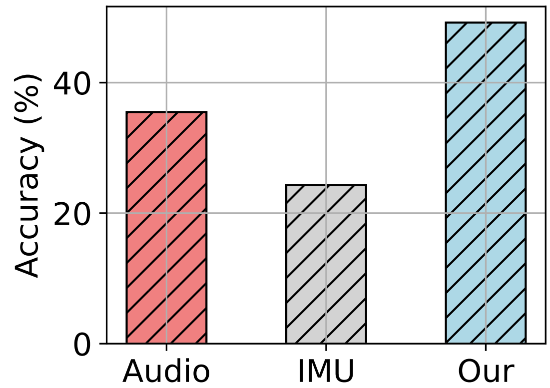


Figure 4.19: *Human activities recognition performance vs. modality.*

three worst activities, which are putting something in a container or on a surface, cutting with scissors, and sleeping. We observe that they are both quiet, and the motion is weakly related to the head, so it is natural to have lower performance.

Modalities fusion. As shown in Figure 4.19, our multi-modal fusion approach (audio+IMU) surpasses the audio-only and IMU-only baselines, improving activity recognition performance by more than 10%.

Table 4.1: *Ablation study on activity recognition for different datasets, the metric is accuracy.*

Model	Ego4D↑	EgoADL↑	SAMOSA↑
Random	0.02	0.02	0.04
ImageBind-FT	0.40	0.51	0.93
Multi-modal	0.401	0.49	0.90
+ Scenario	**0.467**	**0.58**	**0.97**

Ablation study. We conduct an ablation study on different components of our system, including each modality (audio and IMU), multi-modal fusion, scenario as condition, and scenario as condition via Imagebind-embedding, as shown in Tab. 4.1. For reference, we also list the performance of a random guess (indicating the number of classes) and baseline

ImageBind-FT. We observe that

4.5.3 Scenario Recognition

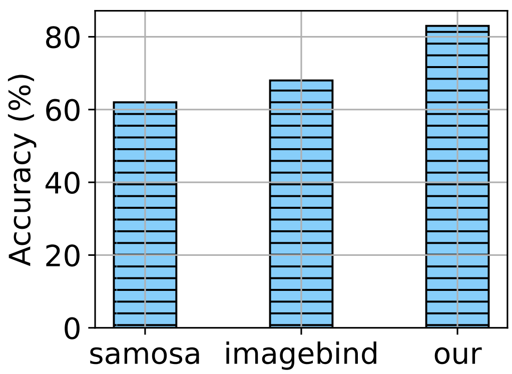


Figure 4.20: *Scenario recognition accuracy on Ego4D.*

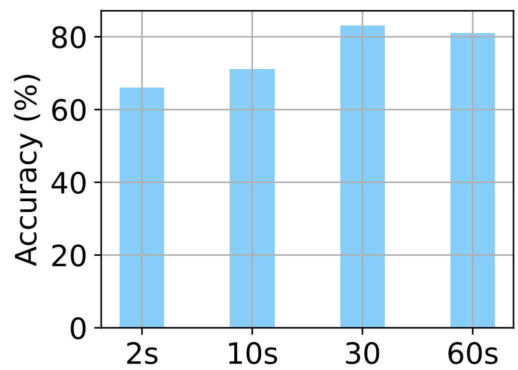


Figure 4.21: *Scenario recognition accuracy vs. data length.*

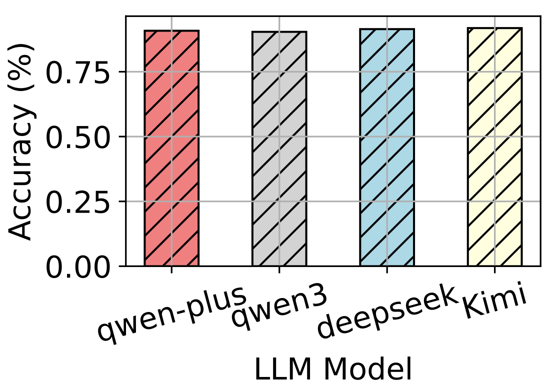


Figure 4.22: *Selection of LLM.*

Model	Ego4D↑
Random	0.01
Multi-modal	0.70
+ Contrastive	0.83
+ Transformer	0.85
+ LLM feedback	0.90

Table 4.2: *Ablation study on scenario recognition.*

Recalled in Sec. 4.4.1, we employ a two-stage model to recognize the scenario by extracting the key-frame. In comparison, we consider audio scene recognition and ImageBind-IMU, which are audio-only and IMU-only methods as the two baselines shown in Fig. 4.20. We observe that our method outperforms the two baselines by 15%, indicating that our long-time multi-modal fusion works well for scenario recognition.

Data length. Based on our earlier definitions, scenarios that require capturing long-term information should use longer data lengths. We present the performance of scenario recognition along with various data lengths in Fig. 4.21. We observe that the ideal data length is set to 30 seconds since the larger input data can reduce the performance.

Selection of LLM. Since our design is compatible with any LLM, we evaluated the performance using different LLMs from various sources. As shown in Fig. 4.22, we observe that the performance is similar across them, indicating that common LLMs are sufficient for our design. For better deployment, we plan to explore the use of smaller LLMs in the future to reduce costs.

Ablation study. We conducted an ablation study to evaluate the contribution of different components in our system, including contrastive learning, the transformer module, and the addition of LLM feedback. The results are presented in Tab. 4.2. For reference, we also include the performance of a random guess (based on the number of classes).

Our observations indicate that contrastive learning is the most effective component, providing the largest performance improvement (13%). The LLM feedback, facilitated by edge-cloud collaboration, yields a further, albeit smaller, gain. Given that LLM feedback is less susceptible to cross-domain variations, we consider this slight improvement to be meaningful.

4.5.4 Spatial Understanding for Scenario Recognition

In Fig. 4.23, we compare the baseline (without motion compensation) with our proposed method across different devices. We report the accuracy for both direction (Dir) and distance (Dist) separately. The results show that our method improves accuracy by approximately 10%when using IMU data for motion compensation.

Regarding performance across devices, we observe a consistent improvement from binaural earphones to 4-channel headphones and further to 4-channel smart glasses, which

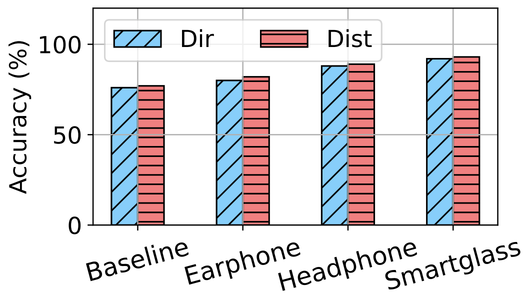


Figure 4.23: *Sound localization accuracy.*

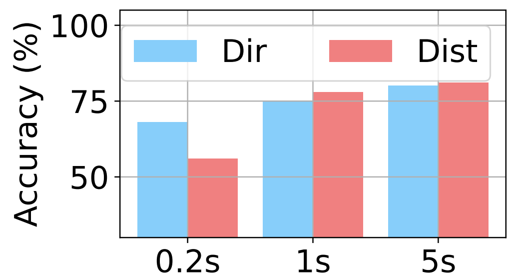


Figure 4.24: *Sound localization accuracy vs. data length.*

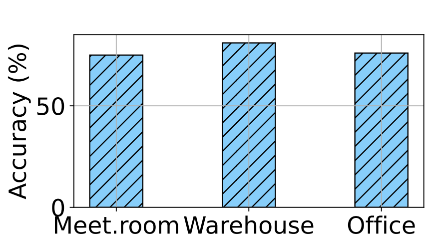


Figure 4.25: *Sound localization accuracy at different places.*

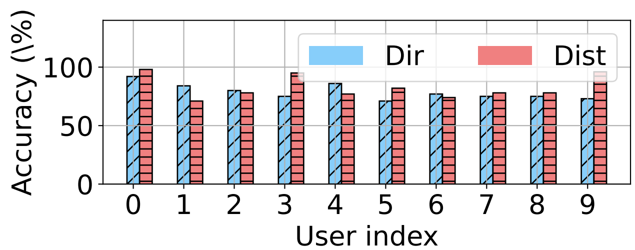


Figure 4.26: *Binaural sound localization accuracy for different users.*

aligns with our expectations. In the following section, we will focus on evaluating data from earphones, which exhibit the lowest performance.

Data length. Since the length of the audio significantly influences the performance of sound localization, we test the accuracy with different length of data in Fig. 4.24. According to the result, we set the data length of sound localization to one second to balance the latency and accuracy.

Different environments. We conducted further assessments of localization accuracy across various environments, such as distinct rooms. Room layout and object configurations induce varying room impulse responses, affecting recordings in conjunction with HRTFs. Our performance evaluation across different rooms is presented in Figure 4.25, demonstrating that our model maintains consistent performance across diverse settings.

Different users. Binaural sound localization depends on individual-specific HRTFs, rendering neural networks calibrated for one user ineffective for others. We assess personalized models across a group of users and present the overall accuracy in Figure 21. Despite performance variations among users, consistent accuracy (>75%) is maintained within the group

4.5.5 User Study

To evaluate the benefits of EgoLog from a user perspective, we conducted a study where volunteers provided subjective ratings of the system's outputs. Recall that our goal is to provide users with a daily log that comprehensively summarizes and represents their activities through two complementary types of output: scenarios and activities. Our final solution integrates both outputs: the daily log displays two rows, with the first row showing scenario information and the second row presenting activity data. Since scenarios are recognized every 30 seconds, the output can contain errors or inconsistencies, even within the same scenario. To mitigate this, we represent each scenario using the normalized probability calculated from the most recent 10 minutes of data. For activities, due to the larger volume of data, we present the results using a bar chart with activity names labeled at the bottom. Users were asked to rate the outputs based on three criteria: Correctness, Information-richness, and User-friendliness. To assess correctness, they were provided with ground-truth data in the same format. The average ratings (on a scale of 1-min to 5-max) for the three dimensions were 4.5, 4.0, and 4.5, respectively.

4.5.6 Overhead

While our system is not specifically designed for real-time responses, it is crucial to maintain low resource consumption.

Table 4.3: *Model runtime latency.*

Section	Interval	CPU	GPU
Scenario	30 seconds	0.55s	9ms
Spatial	2 seconds	2.5ms	2.1ms
HAR	10 seconds	34ms	2.7ms

Latency. EgoLog comprises three core components: (1) scenario recognition (Sec. 4.4.1), (2) sound localization (Sec. 4.4.2), and (3) HAR (Sec. 4.4.3). As shown in Tab. 4.3, most components are lightweight, enabling real-time performance on both CPU and GPU platforms. Notably, scenario recognition on CPU requires 0.55 seconds but is executed once every 30 seconds, rendering its latency negligible. While EgoLog does not mandate real-time user feedback, lower latency enhances user experience by minimizing wait times.

Power consumption. To evaluate our power consumption, we locked the screen while continuously recording both audio and IMU data. We then checked the battery level before and after one hour of recording. Using the Pixel 7 as a reference, we observed approximately a 5% drop in power after one hour of recording. Given that there is potential for further optimization, EgoLog shows promise for effective long-term operation.

Data storage. We record audio at a sample rate of 48 kHz, with a 16-bit depth and 2 channels, resulting in a compressed file size of about 170-210 kbps in MP3 format. This equates to around 70 MB of audio data per hour. For motion data, with an update rate set to 10 Hz, the size in CSV format is approximately 1 kB per second, leading to around 3 MB of data per hour. As a result, we can process this data periodically, ensuring that storage requirements remain manageable and do not become overwhelming.

References

2015. [link].

2020. [link].

2022. Bone-conduction ear microphone | shop motorola solutions. https://shop.motorolasolutions.com/bone-conduction-ear-microphone-system/product/PMLN5464A. (Accessed on 08/19/2022).

2022. Ear bone microphone - earhugger®. https://earhugger.com/product/ear-bone-microphone/. (Accessed on 08/19/2022).

2022. What noises cause hearing loss? | nceh | cdc. https://www.cdc.gov/nceh/hearing_loss/what_noises_cause_hearing_loss.html. (Accessed on 03/16/2023).

Sharath Adavanne, Archontis Politis, Joonas Nikunen, and Tuomas Virtanen. 2018. Sound event localization and detection of overlapping sources using convolutional recurrent neural networks. *IEEE Journal of Selected Topics in Signal Processing*, 13(1):34–48.

Mohammad Arif Ul Alam and Nirmalya Roy. 2014. Gesmart: A gestural activity recognition model for predicting behavioral health. In *2014 International Conference on Smart Computing*, pages 193–200. IEEE.

S Abhishek Anand and Nitesh Saxena. 2018. Speechless: Analyzing the threat to speech privacy from smartphone motion sensors. In *2018 IEEE Symposium on Security and Privacy (SP)*, pages 1000–1017. IEEE.

Apple. 2023. Cmheadphonemotionmanager | apple developer documentation. https://developer.apple.com/documentation/coremotion/cmheadphonemotionmanager. (Accessed on 05/16/2023).

Zhongjie Ba, Tianhang Zheng, Xinyu Zhang, Zhan Qin, Baochun Li, Xue Liu, and Kui Ren. 2020. Learning-based practical smartphone eavesdropping with built-in accelerometer. In *NDSS*.

Sarnab Bhattacharya, Rebecca Adaimi, and Edison Thomaz. 2022. Leveraging sound and wrist motion to detect activities of daily living with commodity smartwatches. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 6(2):1–28.

Shing Chan, Yuan Hang, Catherine Tong, Aidan Acquah, Abram Schonfeldt, Jonathan Gershuny, and Aiden Doherty. 2024. Capture-24: A large dataset of wrist-worn activity tracker data collected in the wild for human activity recognition. *Scientific Data*, 11(1):1135.

You Chang, Namkeun Kim, and Stefan Stenfelt. 2016. The development of a whole-head human finite-element model for simulation of the transmission of bone-conducted sound. *The Journal of the Acoustical Society of America*, 140(3):1635–1651.

Ishan Chatterjee, Maruchi Kim, Vivek Jayaram, Shyamnath Gollakota, Ira Kemelmacher, Shwetak Patel, and Steven M Seitz. 2022. Clearbuds: wireless binaural earbuds for learning-based speech enhancement. In *Proceedings of the 20th Annual International Conference on Mobile Systems, Applications and Services*, pages 384–396.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PmLR.

Tuochao Chen, Malek Itani, Sefik Emre Eskimez, Takuya Yoshioka, and Shyamnath Gollakota. 2024. Hearable devices with sound bubbles. *Nature Electronics*, pages 1–12.

Stefania Cristina, Vladimir Despotovic, Rodrigo Pérez-Rodríguez, and Slavisa Aleksic. 2024. Audio-and video-based human activity recognition systems in healthcare. *IEEE Access*, 12:8230–8245.

Alexandre Defossez, Gabriel Synnaeve, and Yossi Adi. 2020. Real time speech enhancement in the waveform domain. *arXiv preprint arXiv:2006.12847*.

Martin Dietz, Lars Liljeryd, Kristofer Kjorling, and Oliver Kunz. 2002. Spectral band replication, a novel approach in audio coding. In *Audio Engineering Society Convention 112*. Audio Engineering Society.

Xuefu Dong, Yifei Chen, Yuuki Nishiyama, Kaoru Sezaki, Yuntao Wang, Ken Christofferson, and Alex Mariakakis. 2024. Rehearsse: Recognizing hidden-in-the-ear silently spelled expressions. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pages 1–16.

Yariv Ephraim and Harry L Van Trees. 1995. A signal subspace approach for speech enhancement. *IEEE Transactions on speech and audio processing*, 3(4):251–266.

Valentin Gabeur, Chen Sun, Karteek Alahari, and Cordelia Schmid. 2020. Multi-modal transformer for video retrieval. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part IV 16*, pages 214–229. Springer.

Ruohan Gao and Kristen Grauman. 2021. Visualvoice: Audio-visual speech separation with cross-modal consistency. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15490–15500. IEEE.

Shruti Gedam and Sanchita Paul. 2021. A review on mental stress detection using wearable sensors and machine learning techniques. *IEEE Access*, 9:84045–84066.

Rohit Girdhar, Alaaeldin El-Nouby, Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. 2023. Imagebind: One embedding space to bind them all. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15180–15190.

Xinyu Gong, Sreyas Mohan, Naina Dhingra, Jean-Charles Bazin, Yilei Li, Zhangyang Wang, and Rakesh Ranjan. 2023. Mmg-ego4d: multimodal generalization in egocentric action recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6481–6491.

Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, et al. 2022. Ego4d: Around the world in 3,000 hours of egocentric video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18995–19012.

Pranav Gupta, Yaesuk Jeong, Jaehoo Choi, Mark Faingold, and F Ayazi. 2018. Precision high-bandwidth out-of-plane accelerometer as contact microphone for body-worn auscultation devices. In *2018 Hilton Head Workshop*, pages 30–33.

Jiaming Han, Kaixiong Gong, Yiyuan Zhang, Jiaqi Wang, Kaipeng Zhang, Dahua Lin, Yu Qiao, Peng Gao, and Xiangyu Yue. 2023a. Onellm: One framework to align all modalities with language. *arXiv preprint arXiv:2312.03700*.

Jiaming Han, Renrui Zhang, Wenqi Shao, Peng Gao, Peng Xu, Han Xiao, Kaipeng Zhang, Chris Liu, Song Wen, Ziyu Guo, et al. 2023b. Imagebind-llm: Multi-modality instruction tuning. *arXiv preprint arXiv:2309.03905*.

Jun Han, Albert Jin Chung, and Patrick Tague. 2017. Pitchln: eavesdropping via intelligible speech reconstruction using non-acoustic sensor fusion. In *Proceedings of the 16th ACM/IEEE International Conference on Information Processing in Sensor Networks*, pages 181–192.

Zengyi Han, Xuefu Dong, Yuuki Nishiyama, and Kaoru Sezaki. 2023c. Headsense: Visual search monitoring and distracted behavior detection for bicycle riders. In *2023 IEEE 24th International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM)*, pages 281–289. IEEE.

Zengyi Han, En Wang, Mohan Yu, Jie Wang, Yuuki Nishiyama, and Kaoru Sezaki. 2025. Headmon+: Domain adaptive head dynamic-based riding maneuver prediction. *IEEE Transactions on Mobile Computing*.

Zengyi Han, Liqiang Xu, Xuefu Dong, Yuuki Nishiyama, and Kaoru Sezaki. 2023d. Headmon: Head dynamics enabled riding maneuver prediction. In *2023 IEEE International Conference on Pervasive Computing and Communications (PerCom)*, pages 22–31. IEEE.

Xiang Hao, Xiangdong Su, Radu Horaud, and Xiaofei Li. 2021. Fullsubnet: A full-band and sub-band fusion model for real-time single-channel speech enhancement. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 6633–6637. IEEE.

Lixing He, Haozheng Hou, Shuyao Shi, Xian Shuai, and Zhenyu Yan. 2023. Towards bone-conducted vibration speech enhancement on head-mounted wearables. In *Proceedings of the 21st Annual International Conference on Mobile Systems, Applications and Services*, pages 14–27.

Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, and Longbo Huang. 2022. Modality competition: What makes joint training of multi-modal network fail in deep learning?(provably). In *International Conference on Machine Learning*, pages 9226–9259. PMLR.

Sijie Ji, Xinzhe Zheng, and Chenshu Wu. 2024. Hargpt: Are llms zero-shot human activity recognizers? *arXiv preprint arXiv:2403.02727*.

Youna Ji, Jun Byun, and Young-cheol Park. 2017. Coherence-based dual-channel noise reduction algorithm in a complex noisy environment. In *INTERSPEECH*, pages 2670–2674.

Prerna Khanna, Tanmay Srivastava, Shijia Pan, Shubham Jain, and Phuc Nguyen. 2021. Jawsense: recognizing unvoiced sound using a low-cost ear-worn system. In *Proceedings of the 22nd International Workshop on Mobile Computing Systems and Applications*, pages 44–49.

Daichi Kitamura, Nobutaka Ono, Hiroshi Sawada, Hirokazu Kameoka, and Hiroshi Saruwatari. 2016. Determined blind source separation unifying independent vector analysis and nonnegative matrix factorization. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 24(9):1626–1641.

Tom Ko, Vijayaditya Peddinti, Daniel Povey, Michael L Seltzer, and Sanjeev Khudanpur. 2017. A study on data augmentation of reverberant speech for robust speech recognition. In *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5220–5224. IEEE.

Daniel Aleksander Krause, Archontis Politis, and Annamaria Mesaros. 2021. Joint direction and proximity classification of overlapping sound events from binaural audio. In *2021 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, pages 331–335. IEEE.

Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, and Thomas Plötz. 2024. Imugpt 2.0: Language-based cross modality transfer for sensor-based human activity recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(3):1–32.

Huining Li, Chenhan Xu, Aditya Singh Rathore, Zhengxiong Li, Hanbin Zhang, Chen Song, Kun Wang, Lu Su, Feng Lin, Kui Ren, et al. 2020. Vocalprint: exploring a resilient and secure voice authentication via mmwave biometric interrogation. In *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*, pages 312–325.

Tiantian Liu, Ming Gao, Feng Lin, Chao Wang, Zhongjie Ba, Jinsong Han, Wenyao Xu, and Kui Ren. 2021. Wavoice: A noise-resistant multi-modal speech recognition system fusing mmwave and audio signals. In *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*, pages 97–110.

He Lixing, Hou Haozheng, Shi Shuyao, Shuai Xian, and Yan Zhenyu. 2023. Code of vibvoice. https://github.com/CUHK-AIoT-Sensing/vibvoice.

Logitech. 2022. Logitech h650e business headset with noise cancelling mic. https://www.logitech.com/en-hk/products/headsets/h650e-business-noise-cancelling.html. (Accessed on 04/13/2022).

Shengzhe Lyu, Yongliang Chen, Di Duan, Renqi Jia, and Weitao Xu. 2024. Earda: Towards accurate and data-efficient earable activity sensing. In *2024 IEEE Coupling of Sensing & Computing in AIoT Systems (CSCAIoT)*, pages 1–7. IEEE.

J MacQueen. 1967. Some methods for classification and analysis of multivariate observations. In *Proceedings of 5-th Berkeley Symposium on Mathematical Statistics and Probability/University of California Press*.

Saif Mahmud, Ke Li, Guilin Hu, Hao Chen, Richard Jin, Ruidong Zhang, François Guimbretière, and Cheng Zhang. 2023. Posesonic: 3d upper body pose estimation through egocentric acoustic sensing on smartglasses. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(3):1–28.

Irene Martín-Morató, Toni Heittola, Annamaria Mesaros, and Tuomas Virtanen. 2021. Low-complexity acoustic scene classification for multi-device audio: Analysis of dcase 2021 challenge systems. *arXiv preprint arXiv:2105.13734*.

Héctor A Cordourier Maruri, Paulo Lopez-Meyer, Jonathan Huang, Willem Marco Beltman, Lama Nachman, and Hong Lu. 2018. V-speech: Noise-robust speech capturing glasses using vibration sensors. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2(4):1–23.

Daniel Michelsanti, Zheng-Hua Tan, Shi-Xiong Zhang, Yong Xu, Meng Yu, Dong Yu, and Jesper Jensen. 2021. An overview of deep-learning-based audio-visual speech enhancement and separation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*.

Chulhong Min, Akhil Mathur, and Fahim Kawsar. 2018. Exploring audio and kinetic sensing on earable devices. In *Proceedings of the 4th ACM Workshop on Wearable Systems and Applications*, pages 5–10.

Vimal Mollyn, Karan Ahuja, Dhruv Verma, Chris Harrison, and Mayank Goel. 2022. Samosa: Sensing activities with motion and subsampled audio. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 6(3):1–19.

Vimal Mollyn, Riku Arakawa, Mayank Goel, Chris Harrison, and Karan Ahuja. 2023. Imuposer: Full-body pose estimation using imus in phones, watches, and earbuds. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pages 1–12.

Seungwhan Moon, Andrea Madotto, Zhaojiang Lin, Aparajita Saraf, Amy Bearman, and Babak Damavandi. 2023. Imu2clip: Language-grounded motion sensor translation with multimodal contrastive learning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13246–13253.

Frederik Nagel and Sascha Disch. 2009. A harmonic bandwidth extension method for audio codecs. In *2009 IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 145–148. IEEE.

Nobuyuki Otsu. 1979. A threshold selection method from gray-level histograms. *IEEE transactions on systems, man, and cybernetics*, 9(1):62–66.

Xiaomin Ouyang and Mani Srivastava. 2024. Llmsense: Harnessing llms for high-level reasoning over spatiotemporal sensor traces. *arXiv preprint arXiv:2403.19857*.

Muhammed Zahid Ozturk, Chenshu Wu, Beibei Wang, and KJ Liu. 2021. Radiomic: Sound sensing via mmwave signals. *arXiv preprint arXiv:2108.03164*.

Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. 2015. Librispeech: an asr corpus based on public domain audio books. In *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 5206–5210. IEEE.

Karol J. Piczak. 2022. ESC: Dataset for Environmental Sound Classification. In *Proceedings of the 23rd Annual ACM Conference on Multimedia*, pages 1015–1018. ACM Press.

Christoph Pörschmann, Tim Lübeck, and Johannes M Arend. 2020. Impact of face masks on voice radiation. *The Journal of the Acoustical Society of America*, 148(6):3663–3670.

Mirco Ravanelli, Titouan Parcollet, Peter Plantinga, Aku Rouhe, Samuele Cornell, Loren Lugosch, Cem Subakan, Nauman Dawalatabad, Abdelwahab Heba, Jianyuan Zhong, et al. 2021. Speechbrain: A general-purpose speech toolkit. *arXiv preprint arXiv:2106.04624*.

N Reimers. 2019. Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*.

Jorge Reyes-Ortiz, Davide Anguita, Luca Oneto, and Xavier Parra. 2015. Smartphone-Based Recognition of Human Activities and Postural Transitions. UCI Machine Learning Repository. DOI: https://doi.org/10.24432/C54G7M.

Antony W Rix, John G Beerends, Michael P Hollier, and Andries P Hekstra. 2001. Perceptual evaluation of speech quality (pesq)-a new method for speech quality assessment of telephone networks and codecs. In *2001 IEEE international conference on acoustics, speech, and signal processing. Proceedings (Cat. No. 01CH37221)*, volume 2, pages 749–752. IEEE.

Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer.

Sriram Sami, Yimin Dai, Sean Rui Xiang Tan, Nirupam Roy, and Jun Han. 2020. Spying with your robot vacuum cleaner: eavesdropping via lidar sensors. In *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*, pages 354–367.

Florian Schmid, Khaled Koutini, and Gerhard Widmer. 2023. Efficient large-scale audio tagging via transformer-to-cnn knowledge distillation. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE.

Ralph Schmidt. 1986. Multiple emitter location and signal parameter estimation. *IEEE transactions on antennas and propagation*, 34(3):276–280.

Bosch Sensortec. 2021. Inertial measurement unit bmi160. https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi160/.

Hyungjoo Seo, Sahil Bhandary Karnoor, and Romit Roy Choudhury. 2022. Ross: Utilizing robotic rotation for audio source separation. *arXiv preprint arXiv:2203.10072*.

Cong Shi, Xiangyu Xu, Tianfang Zhang, Payton Walker, Yi Wu, Jian Liu, Nitesh Saxena, Yingying Chen, and Jiadi Yu. 2021. Face-mic: inferring live speech and speaker identity via subtle facial dynamics captured by ar/vr motion sensors. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*, pages 478–490.

Statista. 2020. Wearable shipments by category 2024. https://www.statista.com/statistics/690731/wearables-worldwide-shipments-by-product-category/. (Accessed on 02/15/2022).

David Strömbäck, Sangxia Huang, and Valentin Radu. 2020. Mm-fit: Multimodal deep learning for automatic exercise logging across sensing devices. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 4(4):1–22.

Weigao Su, Daibo Liu, Taiyuan Zhang, and Hongbo Jiang. 2021. Towards device independent eavesdropping on telephone conversations with built-in accelerometer. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 5(4):1–29.

Cem Subakan, Mirco Ravanelli, Samuele Cornell, Mirko Bronzi, and Jianyuan Zhong. 2021. Attention is all you need in speech separation. In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 21–25. IEEE.

Ke Sun, Chunyu Xia, Xinyu Zhang, Hao Chen, and Charlie Jianzhong Zhang. 2024. Multimodal daily-life logging in free-living environment using non-visual egocentric sensors on a smartphone. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(1):1–32.

Ke Sun and Xinyu Zhang. 2021. Ultrase: single-channel speech enhancement using ultrasound. In *Proceedings of the 27th Annual International Conference on Mobile Computing and Networking*, pages 160–173.

Marco Tagliasacchi, Yunpeng Li, Karolis Misiunas, and Dominik Roblek. 2020. Seanet: A multi-modal speech enhancement network. *arXiv preprint arXiv:2009.02095*.

Catherine Tong, Shyam A Tailor, and Nicholas D Lane. 2020. Are accelerometers for activity recognition a dead-end? In *Proceedings of the 21st international workshop on mobile computing systems and applications*, pages 39–44.

Ivo Trowitzsch, Jalil Taghia, Youssef Kashef, and Klaus Obermayer. 2019. The nigens general sound events database. *arXiv preprint arXiv:1902.08314*.

Department of the State USA. 2022. Everyday conversations: Learning american english | american english. https://americanenglish.state.gov/resources/everyday-conversations-learning-american-english. (Accessed on 02/02/2022).

Heming Wang, Xueliang Zhang, and DeLiang Wang. 2022. Fusing bone-conduction and air-conduction sensors for complex-domain speech enhancement. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30:3134–3143.

Tianshi Wang, Shuochao Yao, Shengzhong Liu, Jinyang Li, Dongxin Liu, Huajie Shao, Ruijie Wang, and Tarek Abdelzaher. 2021. Audio keyword reconstruction from on-device motion sensor signals via neural frequency unfolding. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 5(3):1–29.

Wikipedia. 2020a. Perceptual evaluation of speech quality - wikipedia. https://en.wikipedia.org/wiki/Perceptual_Evaluation_of_Speech_Quality. (Accessed on 12/02/2022).

Wikipedia. 2020b. Voice frequency - wikipedia. https://en.wikipedia.org/wiki/Voice_frequency. (Accessed on 07/12/2022).

Wikipedia. 2022a. Bone conduction - wikipedia. https://en.wikipedia.org/wiki/Bone_conduction. (Accessed on 07/26/2022).

Wikipedia. 2022b. Contact microphone - wikipedia. https://en.wikipedia.org/wiki/Contact_microphone. (Accessed on 07/26/2022).

Sook Young Won and Jonathan Berger. 2005. Estimating transfer function from air to bone conduction using singing voice. In *ICMC*.

Sean UN Wood, Jean Rouat, Stéphane Dupont, and Gueorgui Pironkov. 2017. Blind speech separation and enhancement with gcc-nmf. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 25(4):745–755.

Huatao Xu, Liying Han, Qirui Yang, Mo Li, and Mani Srivastava. 2024a. Penetrative ai: Making llms comprehend the physical world. In *Proceedings of the 25th International Workshop on Mobile Computing Systems and Applications*, pages 1–7.

Huatao Xu, Panrong Tong, Mo Li, and Mani Srivastava. 2024b. Autolife: Automatic life journaling with smartphones and llms. *arXiv preprint arXiv:2412.15714*.

Huatao Xu, Pengfei Zhou, Rui Tan, Mo Li, and Guobin Shen. 2021. Limu-bert: Unleashing the potential of unlabeled data for imu sensing applications. In *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*, pages 220–233.

Bufang Yang, Siyang Jiang, Lilin Xu, Kaiwei Liu, Hai Li, Guoliang Xing, Hongkai Chen, Xiaofan Jiang, and Zhenyu Yan. 2024a. Drhouse: An llm-empowered diagnostic reasoning system through harnessing outcomes from sensor data and expert knowledge. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(4):1–29.

Bufang Yang, Lilin Xu, Liekang Zeng, Kaiwei Liu, Siyang Jiang, Wenrui Lu, Hongkai Chen, Xiaofan Jiang, Guoliang Xing, and Zhenyu Yan. 2025. Contextagent: Context-aware proactive llm agents with open-world sensory perceptions. *arXiv preprint arXiv:2505.14668*.

Qiang Yang and Yuanqing Zheng. 2022. Deepear: Sound localization with binaural microphones. *IEEE Transactions on Mobile Computing*, 23(1):359–375.

Yongjie Yang, Tao Chen, Yujing Huang, Xiuzhen Guo, and Longfei Shangguan. 2024b. Maf: Exploring mobile acoustic field for hand-to-face gesture interactions. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pages 1–20.

Nima Yousefian, John HL Hansen, and Philipos C Loizou. 2014. A hybrid coherence model for noise reduction in reverberant environments. *IEEE Signal Processing Letters*, 22(3):279–282.

Qian Zhang, Dong Wang, Run Zhao, Yinggang Yu, and Junjie Shen. 2021. Sensing to hear: Speech enhancement for mobile devices using acoustic signals. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 5(3):1–30.

Renrui Zhang, Jiaming Han, Chris Liu, Peng Gao, Aojun Zhou, Xiangfei Hu, Shilin Yan, Pan Lu, Hongsheng Li, and Yu Qiao. 2024a. Llama-adapter: Efficient fine-tuning of language models with zero-init attention.

Siyuan Zhang and Xiaofei Li. 2021. Microphone array generalization for multichannel narrowband deep speech enhancement. *arXiv preprint arXiv:2107.12601*.

Xiyuxing Zhang, Yuntao Wang, Yuxuan Han, Chen Liang, Ishan Chatterjee, Jiankai Tang, Xin Yi, Shwetak Patel, and Yuanchun Shi. 2024b. The earsavas dataset: Enabling subject-aware vocal activity sensing on earables. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 8(2):1–26.

Hongzi Zhu, Yuxiao Zhang, Zifan Liu, Xiao Wang, Shan Chang, and Yingying Chen. 2021. Localizing acoustic objects on a single phone. *IEEE/ACM Transactions on Networking*, 29(5):2170–2183.

Wentao Zhu, Xiaoxuan Ma, Zhaoyang Liu, Libin Liu, Wayne Wu, and Yizhou Wang. 2023. Motionbert: A unified perspective on learning human motion representations. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 15085–15099.